B.Comp. Dissertation

AI in financial data analytics and trading

Budgeted event screening for structural arbitrage

By

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Abstract

This dissertation studies whether machine learning can choose a few prediction-market events to monitor for structural arbitrage during the next hour. A frozen Dome dataset contains 46 complete-outcome events and five ladders of nested conditions. Models are evaluated across thirteen weekly blocks using only earlier training data. Unknown outcomes remain distinct from hours with no opportunity, and capture is evaluated separately from simulated execution. With three slots, a fixed book-feature random forest retains 94.71% of known positive hours and 99.53% of quoted value, versus 73.54% and 60.76% for ranking current quoted margins. Historical event averages already retain 98.41% of pooled value, but 57.14% when events receive equal weight, compared with the forest's 87.06%. Feature, subject-transfer, training-history and resource comparisons show where the result changes. More complex models do not consistently improve capture. Execution delays create partial-basket losses, and excluding December Fed, which supplies 83.15% of quoted value, removes the forest's simulated profit advantage over quoted gap. Dome's API retirement prevented expansion through the original source. The findings support screening within this cohort, with broader validation limited by coverage, concentration, feedback and execution assumptions.

Subject descriptors: Machine learning; Simulation evaluation; Financial markets.

Keywords: prediction markets; structural arbitrage; event ranking; order-book replay; tabular learning.

Implementation: Python 3.12; pandas; scikit-learn; XGBoost; PyTorch/TabM; local parquet data. CUDA acceleration is optional.

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1 Introduction

1.1 Motivation

Prediction markets turn uncertain outcomes into tradable contracts. A YES share pays $1 if its condition is met; a NO share pays when it is not. Related contracts can also impose constraints on their combined payoff. If exactly one of several outcomes must occur, buying one YES share for every outcome guarantees a $1 payoff. A combined purchase cost below $1 creates a potential margin before execution costs. Such opportunities follow from the relationship between contracts, without forecasting the winner.

Finding a displayed gap is only part of the problem. A trading system must repeatedly inspect order books, which record prices and quantities available to buy or sell, check the required combinations and maintain order state. This dissertation studies where that monitoring effort should go. Once an hour, a screen examines every eligible event and chooses a small number to monitor during the following hour.

The primary experiment allows three event slots. This budget is an abstraction: the cohort has at most 19 simultaneous candidates and cannot establish a platform-wide computing bottleneck. The question is how much recorded opportunity a restricted allocation preserves. Screening could be useful when relationship checks, execution infrastructure or risk supervision are costly.

The hourly scan still requires current books for all candidates. Its cost and the cost of ranking must therefore be offset by savings during subsequent monitoring. The experiments measure capture, token-book counts and scoring time from stored features. Feed subscriptions, feature construction and total live system cost remain unmeasured.

1.2 Research questions

The central question is whether current order-book features can identify events that will offer structural arbitrage during the next hour. Three monitoring slots form the primary comparison; five slots provide a secondary comparison. The investigation has three parts.

RQ1 asks what machine learning adds beyond simple priorities: current quoted margins, recent books and historical opportunity averages. The comparisons measure both how many positive event-hours are retained and how much of their quoted value is retained. They also examine which features and model choices support the result.

RQ2 asks when that benefit changes. The analyses vary event weighting, training history and monitoring resources, and test transfer between subject categories. A separate procedure chooses a policy using only results from earlier weeks.

RQ3 asks what happens when the selected events are traded in simulation. Execution replay tests delays, incomplete baskets and funding requirements while keeping the monitoring decisions fixed.

Opportunity and execution are evaluated separately throughout. The first asks where a qualifying basket appears in the recorded books. The second asks what the simulator obtains after orders arrive. A high capture rate alone does not establish profitability.

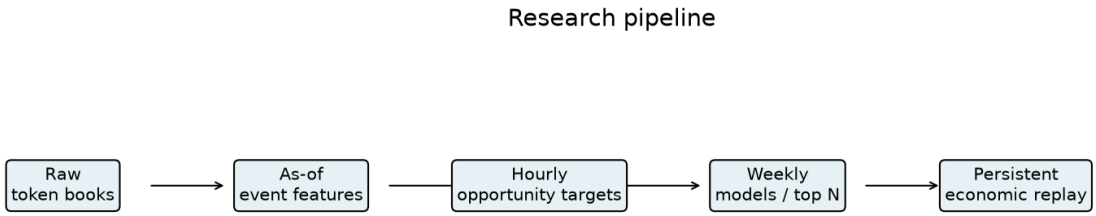


Figure 1. Research pipeline. Opportunity targets and tuning are independent of execution P&L.

1.3 Contributions and scope

The study contributes an event-selection formulation, an implementation and an empirical assessment. Its main safeguards preserve unknown outcomes, let those events compete for monitoring slots, and separate prediction targets from fill assumptions. Its comparisons examine how the apparent benefit changes when large-dollar events, simple historical priorities or available resources receive different emphasis.

The learners are established methods. The contribution lies in adapting and evaluating them for this task. The frozen cohort contains 46 audited complete-outcome events, called negRisk events in the retained metadata, and five ladders of nested conditions. Chapter 3 gives the payoff rules. Source, processed examples and fitted models support reproduction; independently rebuilding raw coverage requires the separately retained token-book archive. Policy choice is retrospective, with no untouched final test.

2 Background and related work

2.1 Forecasting and structural consistency

Wolfers and Zitzewitz (2004) discuss prediction markets as mechanisms that aggregate dispersed information about uncertain outcomes. Their emphasis is the quality and interpretation of market-generated forecasts. This motivates studying these markets but does not establish that every relationship between contracts is continuously arbitrage-free. Forecast accuracy and structural consistency are distinct: an event can be difficult to predict while its contracts still admit a guaranteed-payoff basket.

An event in this study is one ranked object, such as a particular Fed decision. It can contain many binary markets. A token is one YES or NO outcome share and has its own order book. A basket is an audited set of tokens with a known payoff lower bound. A candidate is an event eligible for a scan; the term does not mean an opportunity already exists. A signal is an observed interval in which a basket passes the opportunity test. An attempted order, a completed basket and a profitable executed trade are separate observations.

2.2 Empirical arbitrage research

Saguillo, Ghafouri, Kiffer and Suarez-Tangil (2025) investigate arbitrage on Polymarket, distinguish within-market rebalancing from relationships across markets, and address the cost of searching related conditions using heuristic reduction. Their evidence motivates asking whether mispricing exists rather than assuming a perfectly consistent platform. The task here differs: it predicts which events deserve future monitoring under an explicit slot budget. The cited market-wide profit estimate is neither a label nor a profitability benchmark for this dataset. Historical depth coverage, basket definitions and execution assumptions must be established independently for the present study.

This distinction also limits the role of semantic discovery. A market title suggesting a related subject is insufficient to guarantee compatible settlement rules. The implemented cohort therefore uses frozen audited relationships. Membership is an upstream assumption, whereas the learner ranks events after membership has been fixed. General automatic relationship discovery remains outside the evaluated strategy.

2.3 Monitoring costs and learning from selected events

Trapeznikov and Saligrama (2013) study sequential classification in which a system decides whether another, more costly measurement is worthwhile. Their training data include all sensing modalities, while prediction-time decisions trade acquisition cost against classification error. This supplies a useful principle: the benefit of screening should be assessed against the information cost it avoids. The present task allocates future event monitoring rather than additional measurements for one classification. Event slots and token counts make that allocation explicit, but do not inherit the paper's measured-cost interpretation. Cached-feature timing can quantify only part of the overhead.

Chen, Wang and Yuan (2013) study combinatorial bandits, where a learner chooses a subset and observes outcomes associated with played arms. Their formulation connects resource allocation to the information obtained for future choices. That feedback distinction matters here: historical-label baselines and the earlier-only chooser use outcomes from unselected events as well. They are full-feedback offline comparisons. Their results do not demonstrate a deployed learner that acquires its training labels through three selected events, and this dissertation makes no bandit-regret or exploration guarantee.

2.4 Tabular learning and incremental predictive information

Breiman (2001) introduces random forests as ensembles of randomized tree predictors. For this application, their ability to learn nonlinear feature relationships provides a comparison with a regularised linear score. The original method does not resolve temporal leakage, class imbalance or resource-constrained evaluation; those require application-specific choices.

Chen and Guestrin (2016) present XGBoost as a scalable tree-boosting system. Here it tests whether sequential tree learning improves upon the forest's averaging approach on the event-selection task. Shared chronological populations make their rankings comparable; the bounded searches examine a limited set of configurations, not each family's best attainable performance.

Grinsztajn, Oyallon and Varoquaux (2022) compare neural and tree methods across tabular benchmarks and examine why trees can remain competitive on medium-sized tables. Their findings justify retaining strong classical controls. However, general tabular benchmarks are not evidence about rare, temporally dependent prediction-market opportunities. A comparison in the present setting is still necessary.

Gorishniy, Kotelnikov and Babenko (2025) introduce TabM, an ICLR 2025 method using parameter-efficient ensembling within a tabular neural model. It supplies a recent neural comparator rather than a generic default multilayer network. This study uses the official implementation and preserves member-wise training losses and prediction averaging. Its

smaller bounded search and three-seed two-stage adaptation are application choices, not a replication of the paper's full benchmark programme. Success on broader benchmarks does not guarantee superiority on this small, concentrated cohort.

These studies justify comparing different ways to learn from tabular features. A separate question is what current books add beyond historical priorities. A model can retain valuable hours by repeatedly prioritising the same historically productive event. Historical frequency and value controls therefore test a different explanation from comparisons between classifier architectures. Feature ablation and family exclusion further ask whether current information and transferable relationships contribute beyond that persistence.

2.5 Evaluation, selection bias and the research gap

Cawley and Talbot (2010) show that model-selection criteria can themselves be overfit, with resulting selection bias in reported performance. This separates two safeguards required here. Training and inner tuning must respect temporal availability at every origin; choosing a policy after inspecting all forward blocks still requires independent subsequent evaluation. Reusing those blocks for additional sensitivity analysis improves diagnosis without creating an untouched test.

Bailey et al. (2017) highlight the risk of selecting strategies because they perform well in historical backtests. This concern applies here because several screening policies are compared and the recommended policy is chosen from their pooled chronological results. Earlier-only training and tuning prevent future observations from entering each fit, but do not independently validate the subsequent choice of the winning policy. This study does not estimate the probability of backtest overfitting; its recommendation therefore requires evaluation on independent future data.

Table 1 summarises how these research areas inform the design. The gap addressed is the empirical connection between anticipating depth-supported opportunities, allocating limited monitoring and evaluating the resulting execution scenario. Existing work supplies the building blocks; this study tests their combination on a fixed audited cohort. Its contribution is correspondingly bounded: evidence of cohort-specific screening, with explicit limits on cost, feedback, coverage and generalisation.

Table 1. Related research and the methodological requirements for this study.

Research	**Question addressed**	**Implication for this study**
Saguillo et al. (2025)	Identify structural mispricing and reduce relationship-search work.	Future monitoring allocation, historical depth coverage and executable fills require separate evaluation.
Trapeznikov and Saligrama (2013)	Choose additional measurements under acquisition costs.	Here the choice is which events to monitor next; event and token counts only approximate acquisition cost.
Chen, Wang and Yuan (2013)	Learn subset choices from the outcomes of played arms.	The historical controls here assume full past feedback; selective-feedback learning and regret are not evaluated.
Grinsztajn et al. (2022); Gorishniy et al. (2025)	Compare tree and neural inductive biases on tabular tasks.	Temporal availability, rare baskets and budgeted capture must be tested in the present domain.
Cawley and Talbot (2010); Bailey et al. (2017)	Explain selection bias when choosing models or strategies from finite historical evidence.	Earlier-only fitting protects each prediction; the final policy recommendation still needs independent evaluation.

3 Problem formulation

This chapter follows one monitoring decision from selection to evaluation, then defines the ranking rule, qualifying baskets and outcome labels precisely.

3.1 A worked hourly decision

Consider a fictional scan at 10:00 with five eligible events, A--E. Every event has usable current books. A policy scores them using information available by 10:00 and ranks them in that order. With three slots, it monitors A, B and C until 11:00. The following-hour outcomes are revealed only afterwards.

Table 2. One fictional hourly allocation at N=3; all numbers illustrate the definitions and are not study results.

Event (rank)	Monitored?	Following-hour label	Peak margin ($)
A (1st)	Yes	Known positive	10.00
B (2nd)	Yes	Known negative	0.00
C (3rd)	Yes	Unknown	Unknown
D (4th)	No	Known positive	2.00
E (5th)	No	Known negative	0.00

Suppose A later offers a qualifying basket with a peak margin of $10, while D offers $2. B and E have complete records showing no opportunity. C's future record is incomplete, so its label is unknown. The policy captures one of two known positive hours: 50% recall. It captures $10 of the $12 known quoted value: 83.33% value capture. Its higher value percentage reflects retaining the larger opportunity while missing the smaller one.

C still uses a monitoring slot. Removing it after learning that its outcome is unknown would give the policy information unavailable at 10:00. Unknown outcomes are excluded from evaluation denominators, not from the selection decision. The $10 is an opportunity weight, not simulated trading profit; execution is assessed separately in Chapter 6. All numbers in this example are illustrative.

3.2 Ranking events at each scan

An eligible event is a candidate for monitoring. Let $\mathcal{C}(t)$ be the candidate set at scan t. For each event e, features $\mathbf{x}(e,t)$ use current and earlier recorded states. A policy assigns score $s(e,t)$ and selects the highest-scoring $\min(N, |\mathcal{C}(t)|)$ events. A fixed event-identifier order resolves equal scores. Future outcomes, including whether they will be known, do not affect eligibility or selection.

The selected set is $S_N(t)$. It contains the available number of events up to N and has the largest total score among sets of that size. Slots are filled even when scores are small or negative: selection authorises monitoring, while the separate opportunity and execution rules determine whether a trade is attempted. Occurrence classifiers rank by an opportunity-probability score; two-stage models rank by estimated hourly peak value.

The monitoring interval is $[t, t+H)$, with H equal to one hour. An opportunity at the scan belongs to this window; one first appearing exactly at $t+H$ belongs to the next. The implementation requires the scan interval and label horizon to match, so overlapping allocations cannot silently exceed the budget.

When $|\mathcal{C}(t)| \leq N$, all candidates are selected. These hours cannot distinguish policies by what they exclude. Competitive-hour recall therefore measures capture specifically when the pool is larger than the budget. Equal event counts also need not imply equal token-book processing.

One hour and N=3 are evaluated design choices. N=3 was chosen after development comparisons to create a selective budget, with N=5 secondary. Neither the optimal rescan interval nor a prospectively fixed budget is established. Other budget curves reuse the saved scores and are exploratory comparisons.

3.3 Baskets with a guaranteed payoff

A basket contains all shares needed for its payoff guarantee. For an audited negRisk event, the outcomes are mutually exclusive and exhaustive: exactly one occurs. Buying one YES share of each outcome pays \$1 in total. Buying q complete shares pays q dollars. All required members must be bought.

A ladder instead uses nested conditions. Let A be broader and B narrower, with $B \Rightarrow A$ under compatible settlement rules. Buying $\mathrm{YES}(A)$ and $\mathrm{NO}(B)$ pays \$1 if neither occurs, \$2 if A occurs without B, and \$1 if both occur. B cannot occur without A. The basket therefore has a \$1 lower-bound payoff; the simulator uses that bound without assuming the additional intermediate payment.

An order book must contain enough shares at the required prices. For a basket with m members, $C_j(q,t)$ is the total ask cost of obtaining q shares of member j, walking through the

displayed price levels. The quoted lower-bound margin is

$$G(q, t) = q - \sum_{j=1}^{m} C_j(q, t). \tag{1}$$

Using only the cheapest ask could overstate this margin when little quantity is available there. Each token provides a recorded minimum order size and price tick. The implementation also uses whole-share quantities and a five-share floor. The minimum basket size is the larger of five and the rounded-up largest member minimum. Whole shares and the five-share floor are study choices; the tick constrains price. There is no upper quantity or cash cap.

Separate orders for the legs require an additional check. Let $L_j(q, t)$ be the worst ask price needed to obtain the requested quantity of member j. A basket qualifies only if

$$q - q \sum_{j=1}^{m} L_j(q, t) > 0. \tag{2}$$

This requires a positive margin even if every leg fills at its submitted limit. A positive margin calculated from the average displayed purchase cost alone would not ensure that property. Any strictly positive margin passing the sizing and limit checks qualifies; there is no additional dollar threshold.

3.4 Positive, negative and unknown outcomes

For each event-hour, $y(e, t)$ is one if a basket qualifies during the next hour and zero if none qualifies. Both labels require enough evidence to decide the whole active window. If any required interval is unknown, the hour remains unknown even if an opportunity was seen elsewhere in it. This gives positive and negative hours the same observation requirement.

The value target $v(e, t)$ is the largest qualifying one-basket margin in that hour. It is zero for a known negative and missing for an unknown hour. Repeated snapshots do not add value. Ladder-pair margins are not summed because pairs can share depth. Summing hourly peaks measures retained opportunity, not cumulative tradable profit.

Some opportunities already exist at the scan; others first appear later. A separate diagnostic reports capture of later onsets. The main task includes both, since monitoring must retain opportunities throughout the following hour.

3.5 Measuring capture

Positive-hour recall is the selected number of known positive event-hours divided by all known positive event-hours. Quoted-value capture is selected known v divided by all known v . These are the 50% and 83.33% measures in Section 3.1. Pooled value adds dollars across events before taking the ratio, so a few large opportunities can dominate it.

Pooled value capture at N=3 is the primary comparison. Ties use captured positive hours, then average precision, as specified for the relevant selection step. Average precision summarises how well scores rank positive examples across thresholds. Precision among selected known hours measures how often monitoring finds an opportunity; unknown-slot share records allocations whose outcomes cannot be evaluated. Token counts describe the associated book burden.

Random selection includes each candidate with probability $\min(N, m)/m$ when m candidates compete. Expected positive and value capture can thus be calculated exactly. Economic replay instead uses 20 fixed-seed random policies, since earlier trades change later liquidity use. Monitor-all is the full-resource reference. Recall is undefined when there are no known positives; value capture is undefined when their value is zero. Neither is reported as zero.

4 Data and example construction

The evaluation periods below refer to overlapping parts of one saved dataset. They are separated here so the changing denominators in later results can be read against a single reference.

4.1 Source, event cohort and evaluation periods

The source is saved Dome token-level L2 data: order books with prices and quantities at multiple levels, accompanied by frozen Polymarket metadata. The archive covers 994 token histories, approximately 1.01 GB, from 14 October 2025 to 3 February 2026 at 08:00 UTC. These 112 days and eight hours do not provide continuous complete records for every event.

Dome's API retired during the research. Its documentation gives 28 April 2026 as the end-of-life date for all APIs (Dome, 2026). Through that source, the study could no longer add event histories or fill missing periods. The event universe therefore remained frozen, constraining its size, diversity and coverage.

The fixed list has 51 supported events: 46 negRisk and five ladders. The complete prepared table contains 29,308 eligible event-hours, including training history, 23,452 known outcomes and 470 positives. During the main evaluation, 44 events appear as candidates and 12 have known positive hours. An event-hour means one eligible event at one hourly scan; repeated rows are not independent events.

Table 3. The three evaluation periods used in the report; the periods overlap and their row counts must not be added.

Analysis period	Dates (UTC; end excluded)	Known / eligible event-hours	Positive hours
Main evaluation	11 Nov 2025 to 3 Feb 2026, 08:00	15,787 / 19,766	359
Earlier evaluation	11 Nov 2025 to 8 Jan 2026, 00:00	14,162 / 17,293	343
After warm-up	9 Dec 2025 to 3 Feb 2026, 08:00	7,198 / 8,584	77

The main evaluation supplies the headline 359-positive comparison. The earlier evaluation ends at 8 January because the retained feature table ends on 7 January; additional refits use its 343 positives. The 77-positive period after warm-up evaluates policy choice using earlier weeks. These are overlapping views, not three independent datasets. The earlier feature table

contains 48 events including training history; some do not enter the main evaluation.

Table 4. Forward ranking population.

Measure	Value
Scheduled hours	2,024
Event hours	19,766
Events	44
Known	15,787
Unknown	3,979
Positive	359
Positive events	12
Median candidates	9.00
Maximum candidates	19
Competitive hours at N=3	1,887
Total peak value	8,838.31

Basket membership is fixed in a manifest of 348 definitions: 46 complete sets and 302 ladder pairs. The 302 pairs belong to only five events and often share tokens. Complete sets require audited exhaustive, mutually exclusive outcomes. Ladders require compatible resolution sources, periods and threshold or deadline rules. Freezing membership keeps relationship discovery separate from ranking; it does not reconstruct the platform catalogue available at every past time.

Table 5. Frozen basket membership by structural type.

Structure	Events	Baskets	Token books
negRisk	46	46	872
Ladder	5	302	122

The ladders concern November Bitcoin and Ethereum thresholds, Gensyn sale commitments, Monad launch valuation and shutdown deadlines. Within one audited monthly reach-price family, reaching a higher threshold implies reaching a lower one. The basket buys lower-threshold YES and higher-threshold NO. Different measurement periods or reach/dip rules cannot be combined just because titles mention the same asset. The small ladder sample

reflects available histories and audited relationships, not the market-wide frequency of ladder arbitrage.

The inventory records required tokens, first/last timestamps, file sizes and file fingerprints. Metadata identify member markets and settlement descriptions. There is no complete log of rejected or unavailable platform events, so a market-wide sampling fraction cannot be calculated.

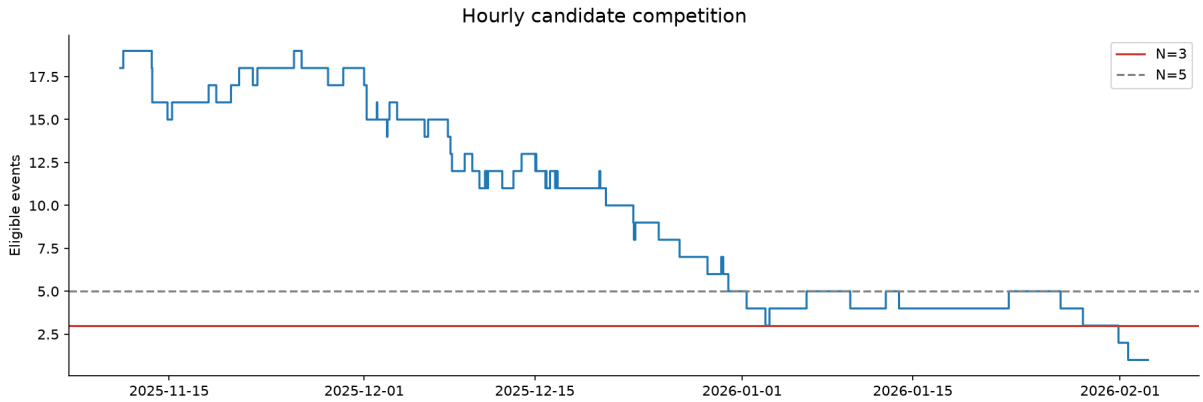


Figure 2. Hourly candidate competition. Candidate counts use the complete eligible pool including unknown labels.

4.2 Constructing a book state at the scan

Each raw row contains source and provider-index times, ask and bid levels, minimum order size and tick size. Prices and sizes must be finite and valid. An empty side is an observation of no displayed liquidity; a missing or malformed side is unknown. A crossed book, whose best bid exceeds its ask, does not provide trusted entry evidence.

For each token, the scan uses the latest source-time row at or before the scan. Equal source timestamps retain the last provider-indexed version under stable ordering. This backward lookup is called as-of alignment. Tokens may update at different times, so book age measures how old each retained state is. A later record never supplies an earlier state, and replay stops at each token's audited recording endpoint.

Source time approximates observation through an idealised feed with the declared execution delay. Provider receive-time delays are unavailable. Thus a 100 ms replay does not demonstrate that the collector delivered updates within 100 ms. Keeping an old state without a freshness cutoff also does not prove that its liquidity remained available at the exchange.

4.3 Eligibility and outcome coverage

Eligibility asks whether an event can enter the current decision. It requires valid active membership, previously seen required tokens and a scan within the saved event observation extent. Outcome coverage asks whether the following hour can be labelled. A usable current state can therefore enter ranking even when its future outcome will be unknown. Only training excludes that target.

Coverage is assessed for each active basket interval. Normally every leg must be observed. An observed blocker can sometimes decide the result despite another missing leg: empty asks on one required leg already rule out buying the basket. Otherwise missing member evidence leaves the interval unknown. Positive and negative labels both require the entire active hour to be decidable.

For example, suppose all legs are observable from 10:00 to 10:40, but a required leg becomes unknown until 11:00. The hour remains unknown even if a qualifying basket appeared at 10:20. Coverage therefore concerns whether the basket can be evaluated, not just the fraction of tokens with records.

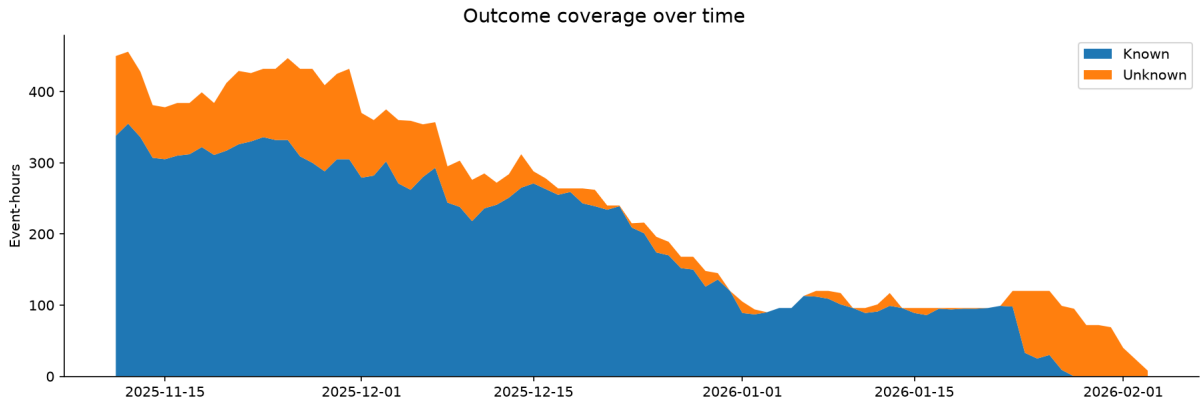


Figure 3. Outcome coverage over time. Unknown intervals do not become negative examples.

Figure 3 counts eligible event-hours by UTC day. Blue includes known positives and known negatives; orange counts unknown labels. Their sum is the candidate population. The falling total reflects fewer eligible event-hours in the fixed cohort. From 28 January, none has a known label although some carried book states still support features. That is loss of evaluable evidence, not evidence that opportunities stopped occurring.

4.4 Book features and metadata

Eleven summaries describe the event at the scan. They count active baskets and tokens, describe valid books and spreads, total displayed ask depth, measure maximum book age, and summarise the margins and size feasibility of its baskets. Table 6 defines each quantity and unit.

Table 6. Current book features before adding hourly lags and changes.

Feature	Definition at scan time	Unit
Active baskets	Number of audited baskets whose validity intervals include the scan.	Count
Active tokens	Number of distinct token books needed by those active baskets; shared legs are counted once.	Count
Seen-member fraction	Previously observed active tokens divided by all active tokens; always one among eligible scans.	0 to 1
Clean-member fraction	Active tokens with nonempty two-sided, non-crossed books divided by all active tokens.	0 to 1
Mean spread	Mean best ask minus best bid across clean active member books; missing if none are clean.	Dollars/share
Maximum spread	Largest best-ask-minus-best-bid difference across clean active members.	Dollars/share
Total ask depth	Sum of displayed ask quantities at all recorded levels across distinct active tokens.	Shares
Maximum book age	Largest elapsed source-time age among the latest observed active member snapshots.	Seconds
Best quoted margin	Maximum guaranteed payoff minus depth-integrated ask cost across supported basket quantities; negative values are retained.	Dollars
Size-feasible fraction	Active baskets with a finite, size- and tick-compatible quoted margin divided by all active baskets.	0 to 1
Positive-margin fraction	Active baskets whose best quoted margin exceeds zero divided by all active baskets; missing margins do not count as positive.	0 to 1

Seen-member fraction is the fraction of active tokens observed previously. Clean-member fraction additionally requires valid, two-sided, non-crossed books. Size-feasible fraction counts active baskets for which a valid depth-based margin can be computed; positive-margin fraction counts those with margin above zero. Eligibility makes current seen-member fraction always one, so that feature cannot distinguish eligible candidates.

Each summary also supplies its previous hourly value and change, giving 33 book features. For current x_t, the additional columns are x_{t-1} and $x_t - x_{t-1}$, with the previous scan exactly

one hour earlier for that event. These are absolute changes. Lags are constructed before eligibility filtering; a missing scheduled scan gives a missing lag. Shares, dollars per share, seconds and total basket dollars remain distinct units. Summed depth is not normalised for event size or computing cost.

The quoted-gap feature is the largest displayed dollar margin across the event's active baskets and supported integer quantities. It subtracts purchase cost from the guaranteed payoff in Section 3.3. For example, 100 complete shares costing $97 give a $3 margin. The feature can be negative when even the best supported combination costs more than its payoff. It omits the additional worst-limit-price check used to qualify an opportunity, so a positive quoted gap alone does not authorise a trade. Missing quoted-gap values rank last in the rule.

Optional metadata add event type, broad subject and days since the earliest frozen member was created. Creation age approximates event maturity, not a verified historical listing date; missing or negative ages remain missing. Subject categories use fixed title rules. Event IDs, winners, realised status, future volume and execution returns are excluded. Frozen descriptions and membership retain a retrospective selection limitation.

Metadata comparisons hold the model settings fixed and add these three inputs, separating added information from tuning. Feature correlations describe associations; the refitted ablations in Chapter 7 examine predictive contribution.

5 Prediction methodology

All policies face the same hourly allocation problem. The comparisons progress from simple current-book rules to learned scores, historical priorities and tests of which information transfers. Exact implementation settings are retained in Appendices C and D.

5.1 Rules and learned policies

A policy is a scoring rule together with the event-selection procedure. Three rules use one current-book summary each: quoted gap prefers a larger margin, depth prefers more displayed ask shares, and freshness prefers a smaller maximum book age. A negative gap can outrank a more negative one because the decision allocates monitoring; a separate test authorises entry.

Logistic regression learns a regularised linear score. Random forest combines trees that capture nonlinear relationships, while XGBoost builds trees sequentially to improve earlier predictions. Each has three variants: fixed book-only, fixed book plus metadata, and tuned book plus metadata. In the tables, book means the 33 summaries and hourly history features; +metadata adds the three event descriptors. Fixed means the settings remain fixed, not that the model is fitted only once. Both fixed and tuned models refit weekly.

Table 7. Fixed classical model settings.

Model	Settings
Logistic classifier	L2 penalty; C=1; balanced class weights; lbfgs solver; maximum 2,000 iterations.
Forest classifier	200 trees; maximum depth 10; minimum leaf size 20; square-root feature subsampling; bootstrap sampling; balanced_subsample class weights.
XGBoost classifier	200 trees; depth 3; learning rate 0.05; minimum child weight 10; row and column subsampling 1; L2 penalty 10; L1 penalty 0; histogram tree construction; negative/positive training-count class-weight ratio.
Two-stage forest	200 trees per stage; depth 8; minimum leaf size 20; all features considered at each split. The probability stage has no class weighting; the amount stage uses squared-error regression on positive values.

Classifiers learn their standard classification losses rather than directly optimising the top-three allocation. Class balancing gives rare positives more weight. Two-stage models instead combine an unweighted occurrence probability with the expected margin conditional on an opportunity:

$$\hat{v}(\mathbf{x}) = \widehat{\mathbb{P}}(y = 1 \mid \mathbf{x})\, \widehat{\mathbb{E}}[v \mid y = 1, \mathbf{x}]. \tag{3}$$

The second stage uses positive-value training rows and clips predicted amounts below zero. An event can thus rank highly because opportunities are likely, valuable when present, or both. The compared two-stage models are a fixed forest and tuned TabM, a neural tabular model. Their scores describe opportunity value, not expected execution profit, and no probability cutoff precedes ranking.

Three equal-weight ensembles combine: the three tuned classifiers; tuned logistic, tuned forest and two-stage TabM; or the two-stage forest and TabM. Each component is converted to percentile ranks within the current candidate pool before averaging. This prevents one model's numerical scale from dominating the combination.

5.2 Preprocessing and model settings

Every learned policy fits preprocessing on its training rows. Classical models replace missing numeric values with training medians, record missingness where applicable, compress magnitudes with a signed logarithm and standardise. Metadata categories become binary indicators. Future examples determine none of these transformations; unseen categories create no new indicators at prediction time.

Each tuned classical model compares eight declared settings. TabM compares four architectures, using training-fitted quantile preprocessing and 32 ensemble members within each stage. Training evaluates each member's loss separately; prediction averages member outputs. Batch size is 1,024, the limit is 60 epochs, and early stopping allows ten epochs without validation improvement. Appendix D retains the transforms, member losses, architecture sizes and scaling rules.

The TabM weekly refit averages three seed-specific products of probability and amount predictions. Its refit duration is the rounded-up median best epoch count from earlier validation fits, retained until the next tuning date. Weekly evaluation outcomes cannot determine when that refit stops. These bounded searches compare specified alternatives, not every model family's attainable best performance.

5.3 Learning from earlier weeks

The first 28 calendar days supply initial training history. Main evaluation runs from 11 November 2025 to 3 February 2026 at 08:00 UTC: 2,024 scans across 13 weekly blocks, the last only eight hours. At each block's start, called its fitting origin in the saved records, models refit on all available earlier training rows and score the next block.

Availability is stricter than having an earlier scan timestamp: a known label's entire one-hour window must end strictly before the fitting date. A window ending exactly at that date is excluded. Earlier evaluation rows can enter later training once this rule is satisfied. All policies still rank unknown candidates; only known rows supply training targets and capture denominators.

Tuning occurs on 11 November, 9 December and 6 January, with at least seven evaluation days remaining. Two validation blocks lie wholly within earlier history. Their models use the same strict label-end cutoff. Settings maximise pooled quoted-value capture at N=3; ties use captured positives, mean validation average precision and finally declared configuration order. Validation also sets TabM stopping durations. The next evaluation block influences none of these choices.

Table 8. Exact chronological validation blocks; endpoints are exclusive.

Tuning date	Block	Validation interval (UTC)	Train rows
11 Nov 2025	1	28 Oct 2025 09:00 to 04 Nov 2025 04:00	3,869
11 Nov 2025	2	04 Nov 2025 04:00 to 10 Nov 2025 23:00	5,707
09 Dec 2025	1	11 Nov 2025 09:00 to 25 Nov 2025 04:00	7,777
09 Dec 2025	2	25 Nov 2025 04:00 to 08 Dec 2025 23:00	12,244
06 Jan 2026	1	25 Nov 2025 09:00 to 16 Dec 2025 04:00	12,312
06 Jan 2026	2	16 Dec 2025 04:00 to 05 Jan 2026 23:00	18,004

Every setting receives the same validation candidates, including unknowns that consume slots. Table 8 gives the actual timestamps; Appendix D explains their indexed-scan construction. Known training rows grow from 7,652 to 23,452, but the last known training scan is on 27 January because later labels are unavailable.

Table 9. Chronological feature, label, fitting and selection procedure.

Step	Operation
1	At each scan, identify valid active baskets and require all active members to have been seen. Keep candidates regardless of later label availability.
2	Read each token at its latest source timestamp at or before the scan; construct current features, consecutive-hour lags and changes.
3	Offline, label the following hour only when every active interval is decidable. Record occurrence and the maximum qualifying one-basket margin; preserve unknown targets.
4	At a weekly origin, fit preprocessing and models on known examples with label-end time strictly before the origin. Retune only at declared origins using earlier inner blocks.
5	Score every eligible event in the evaluation block at its hourly scan. Select up to N by descending score, resolving ties by event identifier.
6	Evaluate selected rows against their known labels. Separately replay selected monitoring windows under fixed arrival and unwind assumptions.

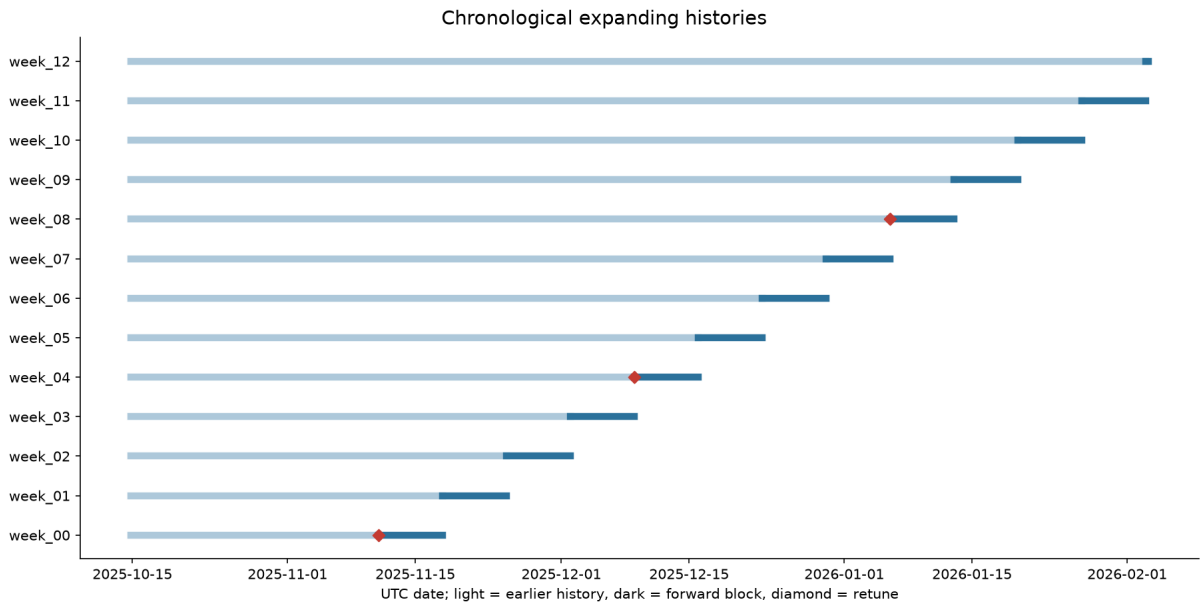


Figure 4. Chronological expanding histories. Full target windows are purged before each origin. Final block is eight hours.

5.4 Policy recommendation and uncertainty

The primary recommendation uses the highest pooled value capture at N=3 across the main evaluation, followed by the tie criteria in Section 3.5. Economic profit and loss (P&L) is assessed afterwards and does not change that criterion. Each prediction uses earlier training, but choosing a winner after comparing all blocks remains retrospective. The later analyses diagnose that same inspected dataset and do not create an untouched final test.

Uncertainty intervals use paired resampling: both policies are compared on the same resampled observations. The main procedure draws the 13 weekly blocks with replacement 10,000 times and reports a 95% interval for the difference. Whole blocks preserve dependence within a week, including the eight-hour final block's actual size. Dependence across weeks can remain.

These intervals hold the fitted models and cohort fixed. They exclude uncertainty from refitting, universe selection and missing outcomes, and do not adjust for all policy comparisons. Event- and family-based resampling below examine different dependence units. Section 8.2 brings these shared limits together; each result also identifies its particular restriction.

5.5 Which features contribute?

Feature ablation means refitting after retaining or removing a specified input group. It tests whether that information improves the recorded ranking while holding other experimental choices fixed. Six views use all 33 book features; the 11 current summaries; current margin alone; all except the quoted-margin group; all except book age; or current margin and age together.

The quoted-margin group contains six current, lagged and changed margin and positive-margin-fraction columns, leaving 27 when removed. Other quote-related information, including size feasibility, remains. Removing age leaves 30. These are targeted comparisons, not removal of every arbitrage-related input.

New fitting uses the earlier evaluation in Table 3. Its retained feature table contains 26,835 rows including training history; 17,293 evaluation rows exactly match saved keys, labels, values and known flags. The full later feature table is unavailable for these additional fits. Full-period analyses can still reuse the original 336,022 saved rows: 17 policies scored on the same 19,766 candidates.

Every view uses fixed forest settings and three seeds, with matched training rows, test candidates and strict cutoffs. Nine weekly origins, the last truncated to two days, give 162 fits. Preprocessing is refitted within each view. Reported seed minima and maxima describe variation across those fits, not confidence intervals for future events. Recomputed N=3 and

N=5 metrics must agree with the original results before further analysis.

5.6 Historical priorities, subject transfer and equal weighting

Historical controls ask whether productive events can simply be prioritised again. Four scores use past event opportunity frequency, past event mean quoted value, last-known event value, or past subject-family mean value. A subject family groups related topics, such as central-bank decisions or elections. It is distinct from a model family such as forests or neural networks.

Only known label windows ending strictly before the current scan enter these scores. Known zeros contribute; unknown windows do not. Event means are smoothed towards family means, and family means towards the global mean, using a fixed 24-observation prior at each level. This reduces reliance on a few observations. An event without history falls back to its family estimate; empty global history gives zero. Last-known value includes observed zeros and uses the family mean only before the first known event outcome. Appendix D gives the exact calculation.

The history timeline has 29,308 deduplicated rows, combining retained earlier examples with later saved outcomes. All controls score the main evaluation. They use event identity explicitly and update hourly; book-feature models omit event identity and refit weekly. Full-feedback means that outcomes from unselected events are also available. The offline records provide that feedback; its acquisition by a deployed three-slot policy is untested and uncosted.

The transfer test removes every training row from one subject family, then fits the fixed forest and logistic model at each earlier weekly origin. Forest uses three seeds and logistic one. Every model ranks the whole ordinary pool before capture is measured within the withheld family; no slots are reserved for it. Matched unrestricted forests come from the ablation fits. Of 81 family-week combinations, 63 contain target-family candidates, giving 189 forest and 63 logistic fits. Missing target-family blocks need no fit.

Each withheld family is a separate scenario. Combining their selections would not produce one three-slot policy. The test examines transfer between categories in this cohort, not performance on independently collected future markets.

Equal-event evaluation asks how well an average opportunity-bearing event is served. First calculate each positive event's capture fraction, then average the 12 fractions equally. Equal-family evaluation first combines outcomes within each of eight positive families, then averages those fractions equally. Selections remain global N=3. Events with no known positives still compete for slots but have no defined capture fraction to average.

Paired event and family intervals use 10,000 bootstrap draws with seed 20261102. Event resampling draws the 12 positive events. Family resampling draws eight families, retains their positive events and recalculates the mean-event difference. Both interval types thus concern the same event-balanced comparison under different dependence assumptions. They neither refit nor adjust for multiple comparisons, and eight families give limited independent evidence.

5.7 Resource budgets and choosing a policy in advance

Five policies enter these comparisons: book-feature forest, book-feature logistic, quoted gap, freshness and historical event mean value. Their existing scores are reused without new labels or execution replay.

Token-observation caps account for event size. An event costs its recorded active-token count; a token shared by two events counts in each. At each scan, events are considered in score order. A whole event is selected if it fits the remaining cap; an expensive event is skipped and consideration continues. Unknown outcomes use the same resources. The caps are 10, 25, 50 and 100, with 50 the primary comparison and no simultaneous three-event constraint.

This holds the maximum budget fixed, not actual spending. Indivisible events leave different amounts unused, greedy packing is not optimal allocation, and selected sets need not be nested as the cap grows. Tokens approximate monitoring work; subscriptions, scan construction and update rates are not measured.

The earlier-only chooser retains N=3 but chooses which policy to use next week. The first four scored weeks, 11 November to 9 December, supply warm-up because earlier out-of-sample scores are unavailable. At each of the nine later fitting dates, the chooser compares candidates' cumulative captured value on earlier known windows ending strictly before that date. It then uses the best policy for the next block. Ties within the declared tolerance follow historical mean, forest, logistic, quoted gap and freshness. Candidate selections are formed before outcomes are consulted, and their original weekly fitting is unchanged.

Static policies are compared on the same after-warm-up period in Table 3. Evaluating alternative past selections requires full feedback, including unselected events. Paired weekly intervals hold the observed chooser path fixed and do not adjust for its retrospective design.

Recorded decision cases use the earlier evaluation. Known positive hours are grouped by whether forest and historical mean select both, only one or neither. The largest-value hour from each disagreement or joint-miss group is shown; ties use time and event identifier. Unknowns remain in the candidate pool. These deliberately selected examples illustrate decisions without representing typical hours or identifying causes.

Table 10. Principal assumptions, their checks and the remaining limits.

Assumption	Check	Remaining limit
Earlier-only inputs and outcomes	Backward as-of tests, strict fitting cutoffs and per-scan/per-origin availability audits.	Implemented safeguards; source-to-receiver latency is unmeasured.
Monitoring budget represents work	Common token-observation caps supplement equal event counts.	Proxy only; event overlap, update rates and total computing cost are not measured.
Full historical feedback is available	Historical controls and the chooser admit known labels only after their windows end.	Unselected events also need labels; this acquisition requirement is not validated.
Relationships transfer across events	Whole-family exclusions, event-balanced summaries and dominant-event removal.	Mixed transfer; no independent future cohort.
Execution approximates obtainable fills	Persistent replay, partial-basket accounting and 0 to 500 ms arrival scenarios.	Simulated fills; queue effects, bounded funding and live execution remain untested.

5.8 Design sensitivity and scoring overhead

Historical mean value is recomputed with prior strengths of 6, 24 and 72 observations at both smoothing levels. The original 24-observation scores must reproduce exactly. The forest is also fitted with 14-day and 28-day histories, instead of all earlier data. Training scans start at the fitting date minus the declared days and keep the strict label-end cutoff. Nine dates and three seeds give 54 new fits, with existing expanding-history fits as controls. Candidates, preprocessing and model settings are shared.

The sensitivity grid was fixed before execution after earlier results had been inspected. All settings are reported without selecting a replacement primary policy. The same exploratory status applies to the feature, historical, transfer and resource analyses above. Changed coverage rules, basket sizes, horizons or execution costs would require raw reconstruction and are not tested here.

Timing uses 12 equally spaced scans from 6--8 January and that block's saved models. Each policy receives five warm-ups and 30 measured calls per scan. The forest is measured with one and four jobs, preserving its saved scores; logistic and quoted gap are references. Costs include cached-column selection, fitted preprocessing, scoring and N=3 allocation. Loading, fitting, raw reads, feature construction, feeds and historical feedback are excluded.

6 Economic replay

Capture measures whether monitoring retains an opportunity. Replay asks what a simulated order obtains after submission. Keeping these stages separate allows the same selected events to be tested under several execution delays.

6.1 Execution assumptions

Each screening policy's hourly selections stay fixed across scenarios. The primary delay is 100 ms from observing a signal to order arrival, with immediate submission. Alternatives are 0, 50, 250 and 500 ms. These are assumed delays, not measurements of the historical collector or a live trading system.

At a signal, the simulator chooses the supported basket quantity with the largest integrated dollar margin, subject to Section 3.3's depth, size and limit checks. Quantity and capital are uncapped. No minimum signal duration, maximum quote age or random failure probability is added. Every policy and delay has its own liquidity-reservation record, maintained throughout replay.

The frozen cohort is evaluated with zero exchange fees and no per-attempt charge. Saved market flags support that scenario, but are not a historical fee-change ledger. Fee-bearing contracts would require recalculated qualification, sizing and execution costs. Zero delay is an optimistic recorded-state comparison, not an estimate of the exchange's maximum obtainable profit.

6.2 From submission to complete or incomplete baskets

When a selected event produces a new signal, the scheduler fixes the basket, quantity and each leg's price limit using the books then available. Simultaneous baskets are prioritised by quoted margin, then basket identifier. Arrival-time books affect fills only after the submission decision is fixed.

Each token order is simulated as fill-or-kill (FOK): it fills completely if recorded depth supports the full quantity within its limit and satisfies the minimum-size and tick rules; otherwise it is cancelled. Separate legs can have different outcomes. A basket is complete when all fill, partial when some fill, and unfilled when none fill. Missing arrival evidence leaves execution unknown.

A complete basket records the conservative payoff minus its integrated purchase cost. An unfilled basket has zero P&L under the no-charge assumption. For a partial basket, the simulator tries to sell the acquired shares at bid depth one second after arrival. Proceeds minus purchase cost give partial-basket P&L; missing or insufficient exit evidence leaves it

unknown. Buying ten shares at $0.40 and selling at $0.30, for example, loses $1.

Entry uses the signal as observed, never its eventual duration. An opportunity that vanishes quickly can therefore be a positive prediction target yet yield no fill or a partial loss. This is the execution risk the separate layer tests.

6.3 Liquidity reuse, profit and funding

Each event waits until the previous attempt's scheduled unwind time before allowing another. Reservations prevent later scans or overlapping ladder pairs from repeatedly treating the same unchanged ask book as fresh liquidity. They survive model refits and weekly boundaries. The supplied events do not share tokens across schedulers; a cohort with that overlap would need shared reservations across events too.

Complete margins enter P&L at completion and partial results at unwind. Known P&L sums only decidable outcomes; unknown outcomes are counted separately. Drawdown is the fall from the previous cumulative P&L peak, starting at zero. Recognising a guaranteed payoff does not mean settlement cash is available.

The main comparison assumes unlimited funding. A separate no-settlement scenario measures peak cash required if complete baskets return no settlement proceeds during the study. Purchases consume cash, known partial sales return it, and unknown orders keep conservative reservations. This scenario does not measure actual capital or return on investment. Monitor-all observes every eligible event, but its earlier trades can reserve depth needed later; it is therefore not guaranteed to earn the highest P&L.

7 Results

Three findings guide the detailed comparisons. For RQ1, the forest captures more opportunities than current quoted gap, but historical averages already retain most pooled dollars. For RQ2, the benefit changes with event weighting, training history and the resource budget. For RQ3, execution delay and one dominant event materially change the economic interpretation.

Results use the main evaluation unless labelled earlier evaluation or after warm-up. Table 3 maps these periods. Recall counts retained positive hours; value capture weights them by their peak margins.

7.1 How much selection is required?

Most scans require choosing among competing events. The main evaluation has 19,766 event-hours from 44 events: 15,787 known labels, including 359 positives from 12 events, and 3,979 unknowns. The hourly pool has a median of nine events and a maximum of 19. It exceeds three at 1,887 of 2,024 scans.

Every deterministic N=3 policy selects 5,973 event-hours, filling fewer slots when fewer candidates exist. Random selection would capture an expected 22.23% of positive hours and 18.93% of value under these changing pool sizes. Those figures provide a reference for allocation, not a market-wide opportunity rate.

Table 11. Monitoring burden at N=3; shares of all eligible event-hours and token counts.

Policy	Event share (%)	Token share (%)	Unknown slots (%)
Forest: book	30.22	21.28	16.16
Quoted gap	30.22	30.5	28.26
Freshness	30.22	19.26	9.98
Depth	30.22	23.26	21.85

Table 11 shows that the same number of event slots can require different numbers of token books. Unknown windows still use resources. Section 7.11 replaces the event-slot budget with a token cap.

7.2 Learned ranking versus current quoted gap

The forest retains substantially more known future opportunity than ranking events by their current quoted margins. At N=3 it captures 340 of 359 positive hours (94.71%) and 99.53% of quoted value. Quoted gap captures 264 hours (73.54%) and 60.76% of value. Table 12 compares all original policies using the same $8,838.31 value denominator.

Table 12. Opportunity capture at N=3.

Policy	Positive hours captured	Recall (%)	Value capture (%)
Forest: book	340	94.71	99.53
Forest + metadata	333	92.76	99.41
Logistic + metadata	331	92.20	99.20
Classifier ensemble	334	93.04	99.16
Tuned logistic	334	93.04	99.06
TabM two-stage	316	88.02	99.04
Logistic: book	329	91.64	98.81
XGBoost + metadata	321	89.42	98.75
Value ensemble	315	87.74	98.58
Forest two-stage	315	87.74	98.57
Tuned XGBoost	321	89.42	98.40
Logistic / forest / TabM	323	89.97	98.37
Tuned forest	328	91.36	98.02
Freshness	261	72.70	95.21
Depth	182	50.70	91.92
XGBoost: book	323	89.97	82.98
Quoted gap	264	73.54	60.76

The result includes anticipating new opportunities: forest captures 205 of 223 positive hours with no opportunity at the scan. A current margin describes the present, while the target covers the following hour. Other book summaries can help distinguish events whose conditions will change; Section 7.6 tests which inputs support that interpretation.

High recall does not make every selected hour productive. Only 6.79% of selected known hours contain an opportunity. Another 16.16% of selected hours have unknown outcomes. Monitoring is therefore selective attention to rare events, with entry still governed by the separate qualification test.

7.3 What does learning add beyond historical averages?

Historical averages explain much of the pooled dollar result. Ranking by past event mean value captures 98.41%, only 1.12 percentage points below forest's 99.53%. Its positive-hour recall is lower: 79.67% versus 94.71%. Historical frequency captures 81.62% of positives and 92.19% of value; last-known value is weaker on both.

Table 13. Historical-label controls on the full evaluation period, N=3.

Policy	Captured / 359	Recall (%)	Value (%)	Unknown slots (%)
Forest: book	340	94.71	99.53	16.16
Quoted gap	264	73.54	60.76	28.26
Freshness	261	72.7	95.21	9.98
Past event frequency	293	81.62	92.19	12.04
Past event mean value	286	79.67	98.41	12.19
Last known event value	190	52.92	59.89	21.01
Past family mean value	260	72.42	93.72	19.69

The historical-mean control repeatedly favours events that previously offered valuable opportunities. Forest adds 54 captured positive hours, and its advantage is larger when events receive equal weight (Section 7.7). Thus the benefit of learning is clearer for broader opportunity coverage than for the pooled value total alone.

These controls assume full feedback, including outcomes from unselected events. The comparison does not establish the cost of collecting those labels in use, and there is no economic replay for the historical policies.

7.4 Can earlier results choose the next week's policy?

The chooser selects forest at all nine fitting dates after warm-up. On the common later period, it captures 75 of 77 positives (97.40%) and 99.63% of value, exactly matching the static forest. Logistic captures fewer positives but slightly more value; historical mean captures 46 positives and 91.59% of value.

Table 14. Common post-warm-up evaluation at N=3: 9 December 2025 to 3 February 2026, 08:00 UTC.

Policy	Captured / 77	Recall (%)	Value (%)
Earlier-only chooser	75	97.4	99.63
Forest: book	75	97.4	99.63
Logistic: book	72	93.51	99.79
Past event mean value	46	59.74	91.59
Quoted gap	67	87.01	96.99
Freshness	57	74.03	86.3

The procedure can therefore recover forest using earlier forward outcomes. However, warm-up already contains 282 of the full 359 positives and $8,251.84 of the $8,838.31 value. Evaluation has only $586.46 left. It cannot independently confirm the large dollar result seen during warm-up. The cumulative objective also preserves an early lead: there is no policy switch demonstrating adaptation. Full feedback is assumed, and separate candidate execution paths cannot be stitched together to infer chooser P&L.

7.5 Do metadata and more complex models improve capture?

Additional modelling does not consistently improve the strongest fixed control. Metadata changes value capture by +0.39 percentage points for logistic, -0.12 for forest and +15.76 for XGBoost. Table 15 holds model settings fixed, so these differences isolate the added descriptors.

Table 15. Matched metadata effects at N=3, percentage points.

Fixed model	Recall change (pp)	Value change (pp)
Logistic	+0.56	+0.39
Random forest	-1.95	-0.12
XGBoost	-0.56	+15.76

Tuning is similarly mixed. TabM captures 99.04% of value and 88.02% of positives; the classifier ensemble captures 99.16% of value. Neither leads the fixed book-feature forest on the primary criterion. These are results for the declared searches and cohort, not a general ranking of neural and tree methods.

7.6 Which book information supports the result?

The full model's capture is not reproduced by learning from current margin alone. All feature comparisons below use the same earlier evaluation, with 343 positives. Each forest view is refitted with three seeds and fixed settings; rules are unchanged.

Table 16. Earlier-period ablations at N=3; three-seed means and observed seed ranges.

Feature view / rule	Features	Recall: mean [range] (%)	Value: mean [range] (%)
All 33 book features	33	94.46 [94.17, 94.75]	99.53 [99.52, 99.54]
Current features only	11	93.39 [93.00, 93.88]	99.10 [98.81, 99.26]
Quoted margin only	1	74.93 [74.64, 75.22]	61.40 [61.36, 61.42]
Without quoted margin	27	87.07 [86.30, 88.34]	99.30 [99.26, 99.36]
Without book age	30	95.24 [95.04, 95.34]	99.48 [99.40, 99.56]
Current margin and age	2	75.90 [75.80, 76.09]	72.70 [72.65, 72.78]
Quoted-gap rule	1	72.30	60.38
Freshness rule	1	71.43	95.16
Depth rule	1	49.27	91.85

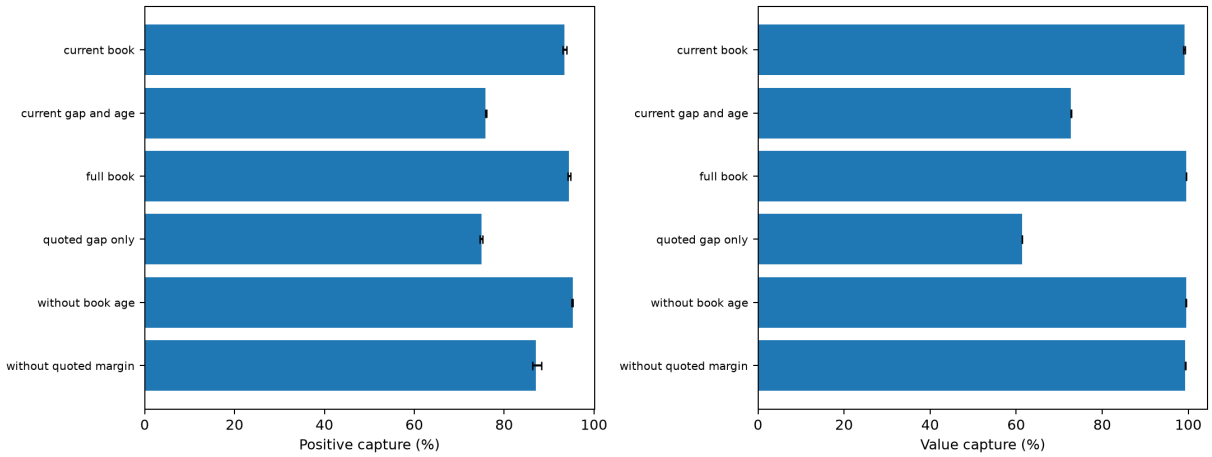


Figure 5. Earlier-period forest feature ablations. 11 November 2025 to 8 January 2026, N=3. Bars show three-seed means; whiskers show seed minima and maxima, not confidence intervals.

The full forest averages 94.46% recall and 99.53% value capture. Current summaries alone reach 93.39% and 99.10%, so the previous-hour values and changes add modest gains of 1.07 and 0.42 percentage points. A margin-only forest reaches 74.93% and 61.40%, close to the

quoted-gap rule's 72.30% and 60.38% on this period. Adding age to margin gives 75.90% and 72.70%. Richer book information matters more than simply applying a nonlinear learner to the gap.

Removing the six margin-group columns lowers recall to 87.07%, while value capture remains 99.30%. Removing the three age columns gives 95.24% recall and 99.48% value. Age therefore does not explain the full advantage in this test. Remaining correlated inputs can substitute for a removed group, so these are tests of predictive dependence rather than causal effects. High value capture can also hide missed low-margin hours.

The original-seed full refit reproduces saved scores to within one trillionth on all 17,293 rows. This checks reproduction, not the original collector's accuracy. Three-seed ranges quantify variability on the same dataset; no reduced view becomes a new primary policy.

7.7 Does the benefit extend across events?

Pooled dollars mainly describe negRisk events. Their 18,095 rows contain 15,177 known labels, 345 positives and $8,835.79 of value. The ladders contain 1,671 rows, 610 known labels, 14 positives and $2.51, about 0.028% of the denominator. Unknown shares are 16.13% for negRisk and 63.49% for ladders.

Table 17. Capture by basket type within the shared N=3 allocation.

Structure / policy	Known positives	Captured	Recall (%)	Value (%)
negRisk: Forest: book	345	331	95.94	99.53
negRisk: Quoted gap	345	252	73.04	60.75
negRisk: Freshness	345	256	74.2	95.22
Ladder: Forest: book	14	9	64.29	93.88
Ladder: Quoted gap	14	12	85.71	96.5
Ladder: Freshness	14	5	35.71	65.2

Forest captures 331 negRisk positives and nine ladder positives; quoted gap captures 252 and 12. The ladder result reverses the pooled ordering, although the small incomplete sample cannot reliably rank learners for ladders generally.

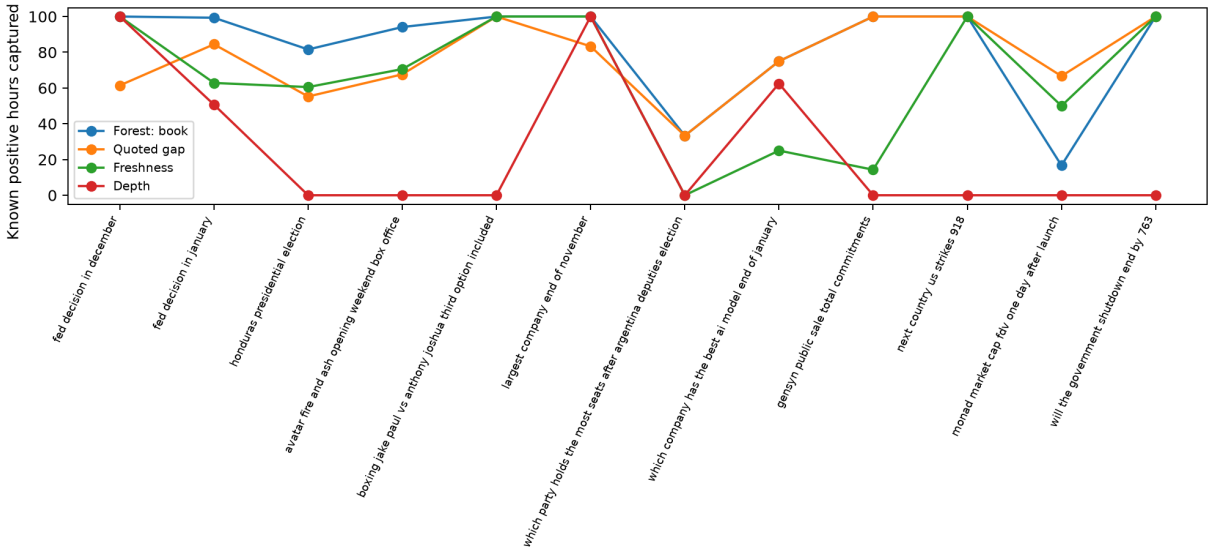


Figure 6. Capture across opportunity-bearing events. N=3; events are ordered by quoted-value contribution. Each event uses its own positive-hour denominator.

Individual events also differ. Both forest and freshness capture all 96 December Fed positives, while quoted gap captures 59. In Honduras, the counts are 31 of 38 for forest, 21 for quoted gap and 23 for freshness. For the Monad ladder, forest captures one of six, quoted gap four and freshness three. Monad's total value is only $0.166, so its misses barely change pooled dollars.

Table 18. Equal-weight capture over 12 positive events and eight positive families, N=3.

Policy	**Event recall (%)**	**Event value (%)**	**Family recall (%)**	**Family value (%)**
Forest: book	83.34	87.06	89.88	97.19
Quoted gap	77.26	81.01	82.49	84.96
Freshness	65.27	73.19	74.01	85.42
Logistic: book	66.56	71.51	71.64	76.09
TabM two-stage	74.36	85.36	78.6	91.99
Past event frequency	39.92	46.68	44.78	50.91
Past event mean value	52.74	57.14	47.15	57.37
Last known event value	47.88	47.36	56.8	58.82
Past family mean value	40.39	41.64	37.6	39.01

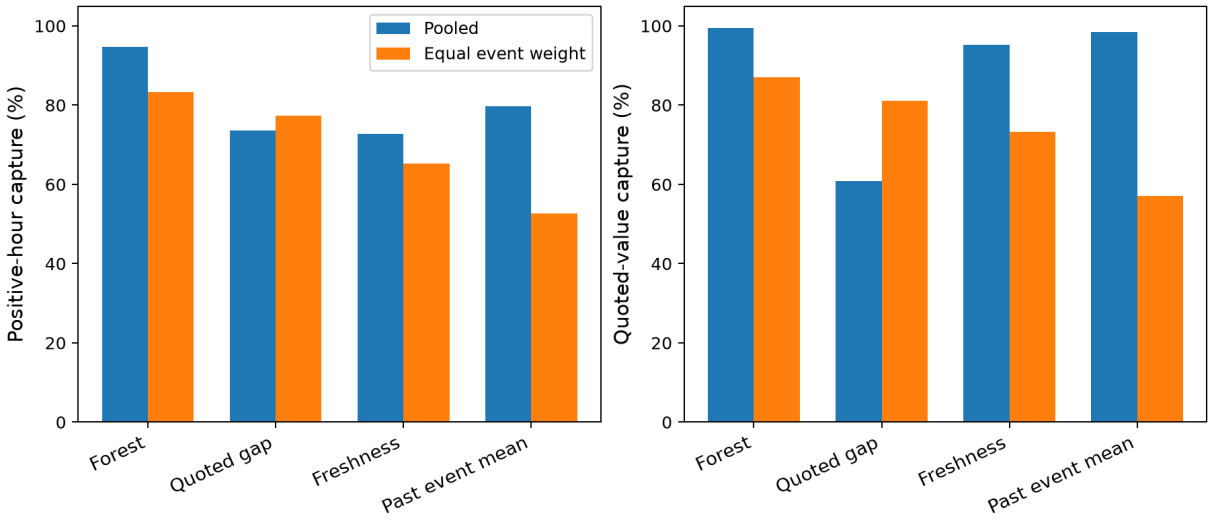


Figure 7. Pooled and event-balanced capture measure different objectives. Full period, unchanged global N=3 selections. Equal-event means use the 12 events with known positive evaluation hours; zero-positive events still compete for slots.

Giving each positive event equal weight makes this distinction visible. Forest averages 83.34% recall and 87.06% value capture across 12 events, versus quoted gap's 77.26% and 81.01%. Its value advantage shrinks from 38.77 pooled points to 6.05 event-balanced points. Historical mean falls from 98.41% pooled value to 57.14% event-balanced value, showing its greater reliance on repeatedly valuable events. Forest has the highest event- and family-balanced point estimates among the 21 original and historical policies.

Table 19. Forest-minus-control differences in mean event value capture; conditional paired intervals.

Comparator	**Difference (pp)**	**Event 95% interval (pp)**	**Family 95% interval (pp)**
Quoted gap	6.05	[-4.98, 16.81]	[-5.75, 16.03]
Freshness	13.87	[-6.36, 33.05]	[-3.96, 33.38]
Logistic: book	15.55	[2.44, 33.22]	[3.07, 35.36]
TabM two-stage	1.7	[-5.74, 9.95]	[-4.10, 8.68]
Past event mean value	29.91	[5.75, 55.76]	[7.05, 57.16]

The uncertainty is wider than the pooled ordering suggests. Forest's mean-event value difference from quoted gap is 6.05 points, with a family-cluster interval of -5.75 to 16.03. It spans zero. Intervals also span zero against freshness (-3.96 to 33.38) and TabM (-4.10 to 8.68), but stay positive against historical mean (7.05 to 57.16) and logistic (3.07 to 35.36). These intervals use eight positive families without adjustment for multiple comparisons.

7.8 Dependence on December Fed and evaluation week

Removing December Fed tests how much the result relies on one unusually valuable event. It supplies 83.15% of quoted value. After removing it from each candidate pool and reranking, forest captures 251 of the remaining 263 positives (95.44%), against quoted gap's 214 (81.37%). Value capture is 97.50% versus 88.36%.

Table 20. Opportunity capture after removing December Fed and reranking at N=3.

Policy	Positive hours captured	Recall (%)	Value capture (%)
Forest: book	251	95.44	97.5
Quoted gap	214	81.37	88.36
Freshness	197	74.9	89.95

Capture therefore remains stronger after exclusion. Section 7.14 shows that the economic advantage does not. This procedure frees the excluded event's possible slot and reranks without retraining; it is not a model trained without that event. Saved outputs repeat the diagnostic for all 12 positive events.

Table 21. Forest-minus-control capture differences and conditional 95% block intervals.

Comparator	Recall difference (pp)	95% interval (pp)	Value difference (pp)	Value 95% interval (pp)
Quoted gap	21.17	[7.10, 30.30]	38.77	[2.33, 60.06]
Freshness	22.01	[16.50, 26.21]	4.32	[1.51, 21.53]
Depth	44.01	[25.33, 60.78]	7.61	[2.24, 40.55]
Logistic: book	3.06	[0.75, 4.75]	0.72	[-0.01, 5.64]
TabM two-stage	6.69	[4.48, 9.68]	0.48	[-0.04, 0.76]

Against freshness, forest's pooled value gain is 4.32 percentage points, with a paired weekly interval of 1.51 to 21.53. Its smaller differences from logistic and TabM span zero. These data do not clearly separate the strongest learned policies under the resampling assumptions in Section 5.4.

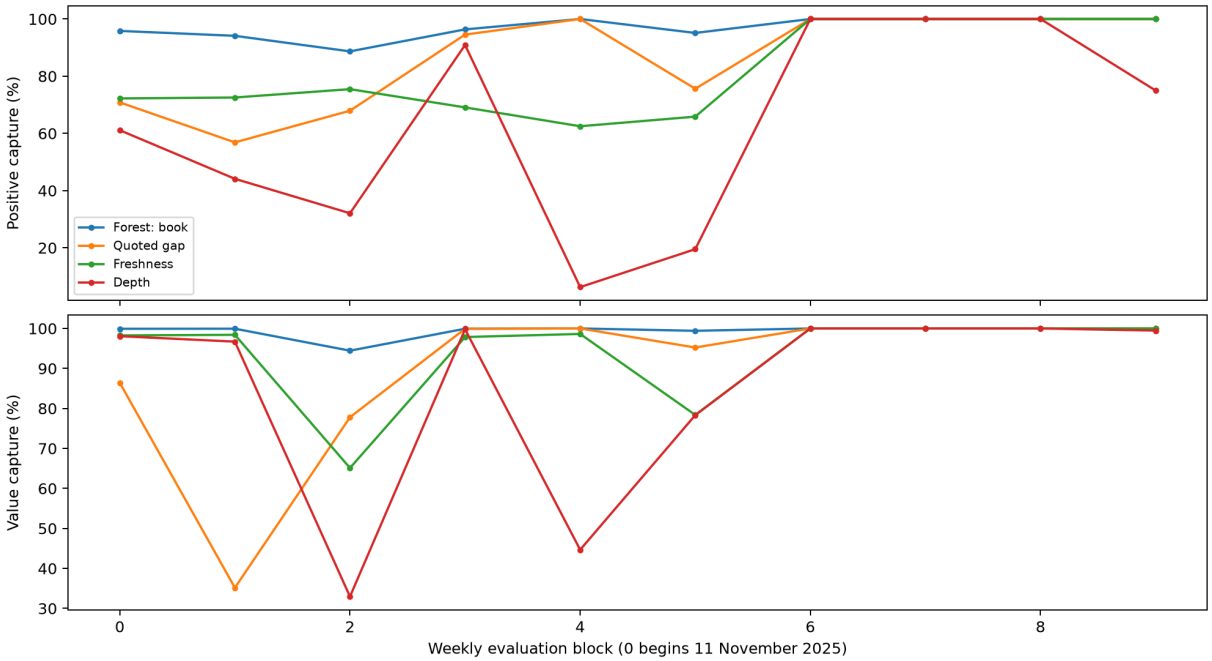


Figure 8. Weekly capture against simple controls. N=3; gaps indicate an undefined denominator rather than zero capture.

Weekly percentages change with candidate competition, positive counts and value concentration. The pooled total weights weeks by their opportunity denominators, not equally. The last eight-hour block has no known labels. It still uses monitoring slots, but its capture is undefined and plotted as a gap. The weekly CSV retains these denominators for inspection.

7.9 Learning without examples from the same subject family

Transfer is uneven. Removing a subject family's training rows tests whether the model can rank that category using relationships learned elsewhere. It still competes against all other candidates. Each family in Table 22 is a separate scenario; RF means random forest, LR logistic regression, and n/a an undefined capture fraction.

Table 22. Earlier-period family transfer at N=3; forest columns are three-seed means.

Target family	Positive hours	Full RF recall (%)	Held RF recall (%)	Held RF value (%)	Held LR recall (%)
Central bank	231	99.57	95.96	98.41	87.01
Crypto	13	58.97	74.36	93.76	15.38
Culture	34	95.10	96.08	99.34	94.12
Elections	41	78.86	67.48	75.56	80.49
Geopolitics	0	n/a	n/a	n/a	n/a
Shutdown	1	100.00	100.00	100.00	0.00
Sports	9	92.59	100.00	100.00	100.00
Technology / business	14	88.10	92.86	96.26	85.71
Tweet counts	0	n/a	n/a	n/a	n/a

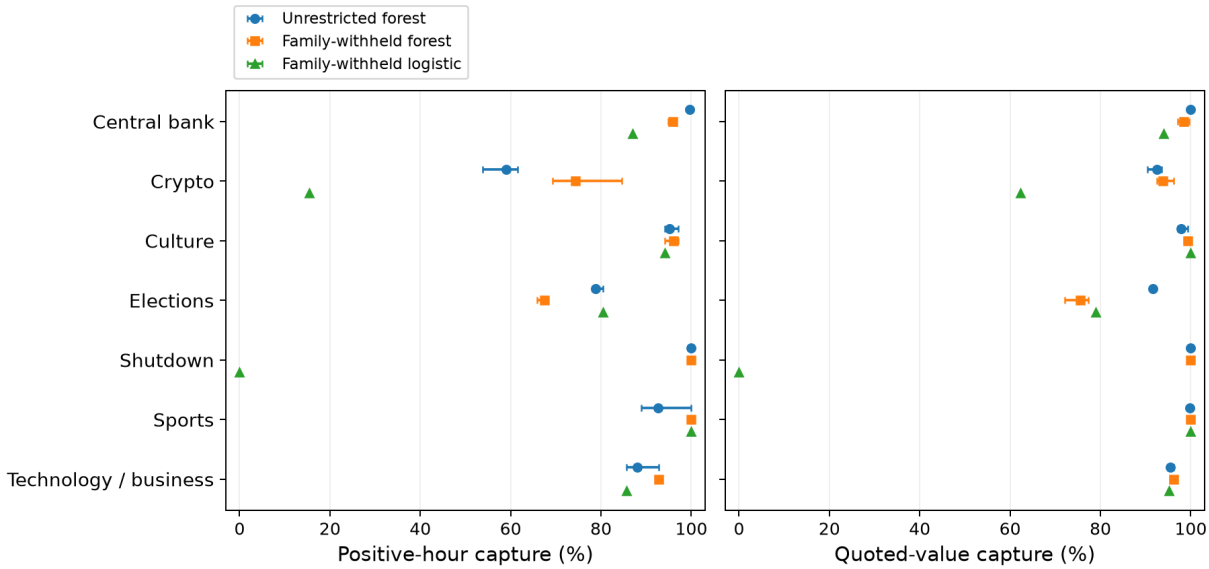


Figure 9. Transfer when a whole family is excluded from fitting. Earlier feature subset, N=3. Each family has its own fitted scenario; selection ranks the whole candidate pool. Forest points show three-seed means and whiskers show seed ranges, not confidence intervals. Zero-positive families remain in the table.

Without central-bank training rows, forest mean recall for that family falls from 99.57% to 95.96%, and value capture from approximately 100% to 98.41%. The three-seed ranges are 94.81--96.54% and 97.08--99.65%. The dominant family's strong capture therefore does not require its own training history here, although it declines from the unrestricted control.

Elections transfer less well. Forest recall drops from 78.86% to 67.48%, and value capture from 91.67% to 75.56%. Withheld logistic reaches 80.49% and 79.01%, exceeding forest on both. Removing crypto, culture, sports or technology/business rows instead improves mean forest recall, while shutdown is unchanged. Such changes can reflect both altered fits and competition for slots.

Small denominators limit the interpretation: shutdown has one positive hour, sports nine and crypto 13. Geopolitics and tweet counts have none in the earlier evaluation, so remain visible with undefined capture. The test concerns these recorded subject groups, not independent future markets.

7.10 Sensitivity to historical smoothing and training length

The historical-baseline result is stable across the declared smoothing choices. Using priors of 6, 24 and 72 observations gives full-period value capture of 98.34--98.56% and recall of 79.39--80.50%. Event-balanced value is 54.44--57.14%, still below forest's 87.06%. Neither the strong pooled baseline nor its weaker equal-event coverage depends on the single original prior.

Table 23. Declared design sensitivities at N=3: historical controls use the full period; forest fits use the earlier feature period. Forest entries are three-seed means [minimum, maximum]; ranges are not confidence intervals.

Setting / population	Recall (%)	Value (%)	Event value (%)
Historical mean, prior 6 / 359 positives	79.39	98.34	57.09
Historical mean, prior 24 / 359 positives	79.67	98.41	57.14
Historical mean, prior 72 / 359 positives	80.50	98.56	54.44
Expanding forest / 343 positives	94.46 [94.17, 94.75]	99.53 [99.52, 99.54]	85.98 [85.54, 86.52]
28-day forest / 343 positives	94.36 [93.88, 95.04]	99.62 [99.42, 99.91]	85.18 [83.15, 86.31]
14-day forest / 343 positives	90.38 [90.09, 90.67]	93.71 [90.81, 95.33]	81.87 [81.46, 82.24]

A month's training history performs similarly to retaining all earlier data. On the earlier evaluation, 28-day forest mean recall is 94.36% and value capture 99.62%, versus expanding

history's 94.46% and 99.53%. Two weeks is weaker: 90.38% recall and 93.71% mean value capture. Its value range of 90.81--95.33% is also wider than expanding history's 99.52--99.54%.

The result identifies sensitivity to aggressive forgetting, not an optimal history length. Changing the window also changes sample size and which events supply training rows. All declared settings remain reported, and the primary policy is unchanged. Historical rows in the table use 359 positives from 12 events; forest rows use the earlier evaluation's 343 from 11. They support within-period comparisons, not matched comparisons across those two periods.

7.11 What changes when more monitoring is available?

Better ranking matters most with a tight budget. At N=5, forest captures 98.05% of positives and 99.99% of value; quoted gap reaches 93.59% and 98.45%. Two extra slots shrink their value gap from 38.77 to 1.54 percentage points.

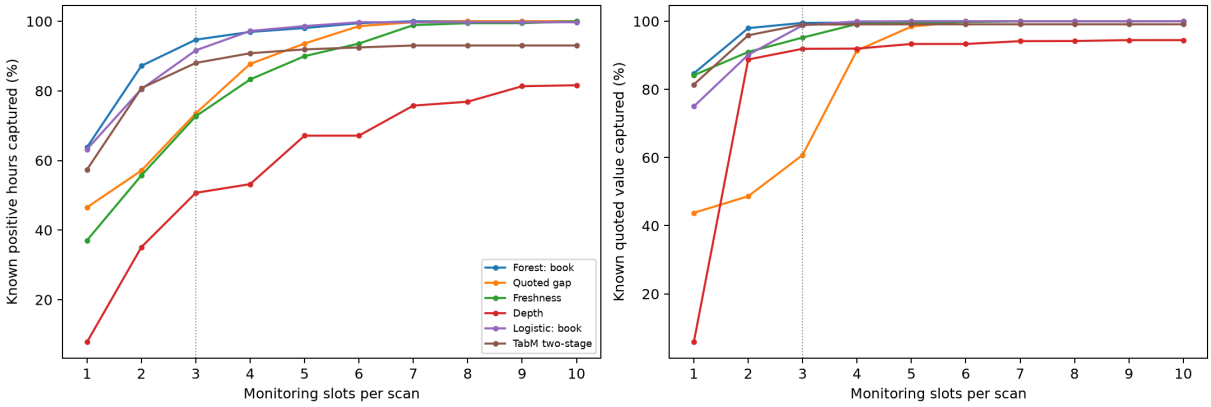


Figure 10. Opportunity capture across monitoring budgets. Full-period saved predictions are reranked; these budgets have not been replayed economically.

At N=1, forest captures 63.79% of positives and 84.71% of value, against quoted gap's 46.52% and 43.78%. With two slots it reaches 87.19% and 97.98%. By ten, forest, quoted gap and freshness all retain every known opportunity. These additional saved-score allocations have no economic replay.

Table 24. Full-period capture with a cap of 50 event-token observations per scan.

Policy	Tokens / scan	Captured / 359	Recall (%)	Value (%)
Forest: book	45.6	354	98.61	99.9975
Logistic: book	44.3	357	99.44	99.9997
Past event mean value	45.91	347	96.66	99.952
Quoted gap	46.19	343	95.54	98.5855
Freshness	44.87	349	97.21	99.8631

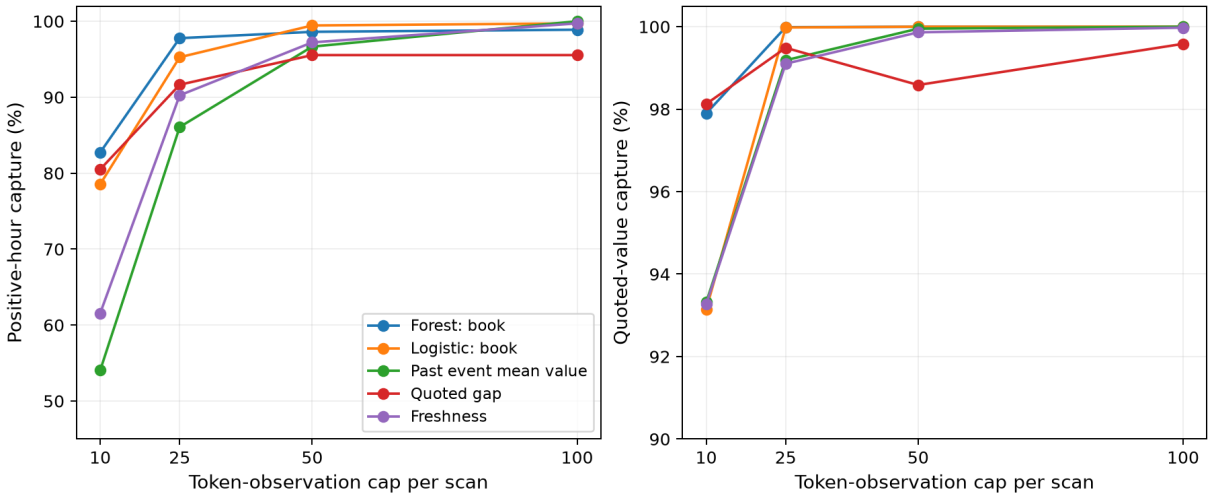


Figure 11. Capture under common monitoring-resource caps. Full period; no three-event cap. Whole events are packed in score order. Shared tokens across events are counted separately. Greedy allocations need not be nested as the cap rises; scanning and inference costs are excluded.

A token cap changes the ordering again. At the primary cap of 50, logistic captures 357 of 359 positives, forest 354 and historical mean 347. Forest's recall difference from logistic is -0.84 points, with a weekly interval of -2.20 to 0.79; against historical mean it is 1.95 points, with an interval of -1.99 to 10.32. Neither establishes a clear forest advantage.

Cap 25 favours forest recall; at cap 100, historical mean retains all positives and value. At cap 10, quoted gap retains slightly more value than forest, which has higher recall. A policy's apparent superiority therefore depends on both the resource constraint and the metric.

At cap 50, forest averages 45.60 token observations and 4.33 events, versus logistic's 44.30 and 4.86. The cap is shared, but spending differs and there is no N=3 restriction. Greedy packing and token counts approximate allocation cost.

7.12 Scoring cost and the savings needed to justify it

Scoring overhead can be measured from stored features even though total monitoring cost is unavailable. The timing sample has four to five candidates per scan from one block, below the main evaluation's maximum 19. Table 25 includes scoring and N=3 selection.

Table 25. Warm cached-feature scoring and N=3 selection: 12 scans, 30 measured calls per scan and policy. The final column divides total measured overhead by avoided event-token observations; scan and feature-construction costs are excluded.

Policy	Median (ms)	95th percentile (ms)	Avoided tokens / scan	Overhead / token (µs)
Forest, 4 jobs	30.838	31.908	55.0	564.35
Forest, 1 job	11.792	12.946	55.0	218.53
Logistic	4.67	5.447	47.08	102.11
Quoted gap	2.424	2.808	36.0	69.63

The four-job forest takes a median 30.84 ms, versus 11.79 ms with one job, 4.67 ms for logistic and 2.42 ms for quoted gap. Both forest settings preserve the predictions. Dispatching parallel work adds overhead on this small sample. These warm, fixed-order timings and their 95th percentiles do not measure live system tail latency; environment details accompany the saved measurements.

The practical question is whether avoided monitoring exceeds screening overhead. Let K be all candidate event-token observations, S the selected observations, A the additional cost of scanning and feature construction, I scoring and allocation cost, and c the average cost of continuing to monitor one token for the next hour. Under this additive approximation, savings require

$$(K - S)c > A + I. \tag{4}$$

The four-job forest avoids a mean 55 observations in the sampled scans. Measured overhead alone requires 564.35 microseconds of saved work per avoided token-hour, or 218.53 with one job. Actual required savings are higher when A adds cost. Token overlap, unequal update rates, feeds and full historical feedback can also change the comparison. The calculation makes the requirement explicit; it does not establish end-to-end savings.

7.13 Examples of correct selections and misses

The recorded cases show both advantages and failures. Of 343 earlier positive hours, forest and historical mean both select 264; forest alone selects 60, historical mean alone six, and neither 13. Forest-only hours contain $135.14 and history-only hours $36.52; most value is retained by both. Table 26 shows the largest-value example from each disagreement or joint-miss group under Section 5.7's rule.

Table 26. Largest known-value examples of selection disagreements and joint misses, earlier subset, N=3.

Case / event	Scan (UTC)	Peak value ($)	Forest rank	History rank
A: Honduras election	24 Nov 2025 21:00	24.55	2	7
B: Honduras election	27 Nov 2025 00:00	33.37	5	3
C: Avatar box office	17 Dec 2025 07:00	1.58	4	8

In Case A, scan margin improves from -$0.18 to -$0.08. Forest ranks the event second and historical mean seventh before a known $24.55 peak. In Case B, the same event's margin worsens from -$0.14 to -$0.24 at a later scan. Forest ranks it fifth while historical mean ranks it third, capturing $33.37 that forest misses. These traces illustrate decisions without attributing them causally to the margin feature.

In Case C, forest ranks the event fourth, immediately below the three-slot cutoff; historical mean ranks it eighth. Both miss $1.58. This is an allocation miss against competing scores, not a failed binary probability threshold. Unknown candidates remain in every pool, and complete features and ranks are saved. The selected cases are illustrations rather than representative hours.

7.14 Do retained opportunities produce higher simulated profit?

Forest earns more than quoted gap in the primary execution scenario, but the advantage depends on one event. Table 27 reports known P&L at 100 ms. An attempt is a submitted basket; completion requires all legs. Random-policy rows average 20 seeds.

Table 27. Economic comparison at N=3 and 100 ms.

Policy	Known P&L ($)	Attempts	Complete baskets
Forest: book	5,873.32	453	436
Forest + metadata	5,873.60	456	439
Logistic + metadata	5,892.81	434	416
Classifier ensemble	5,902.50	462	445
Tuned logistic	5,896.85	444	426
TabM two-stage	5,857.99	431	413
Logistic: book	5,878.47	438	420
XGBoost + metadata	5,915.38	449	433
Value ensemble	5,834.01	428	410
Forest two-stage	5,921.18	454	439
Tuned XGBoost	5,886.61	456	440
Logistic / forest / TabM	5,874.54	437	419
Tuned forest	5,881.06	466	449
Freshness	5,778.40	400	384
Depth	5,745.85	285	280
XGBoost: book	5,187.45	454	437
Quoted gap	4,370.05	368	355
Monitor all	5,944.97	499	481
Random mean (20 seeds)	1,123.15	124.6	119.15

Forest produces $5,873.32 from 453 attempts: 436 complete baskets contribute $6,290.08 and 17 partial baskets -$416.76. Quoted gap produces $4,370.05, monitor-all $5,944.97 and random mean $1,123.15. All attempted forest outcomes are known, although candidate opportunity labels can still be unknown.

Its largest cumulative P&L drawdown is $206.15. The largest basket requests 73,067 shares per leg, and peak required cash reaches $1,426,124.21 in the no-settlement scenario. The

margins therefore rely on substantial position and funding assumptions. They are not returns measured on deployed capital.

The two-stage forest earns the highest model P&L, $5,921.18, while fixed book forest leads opportunity capture. Freshness and depth are economically close to the learned group. The metric used to recommend a monitoring policy therefore matters: the study retains opportunity capture as its declared criterion.

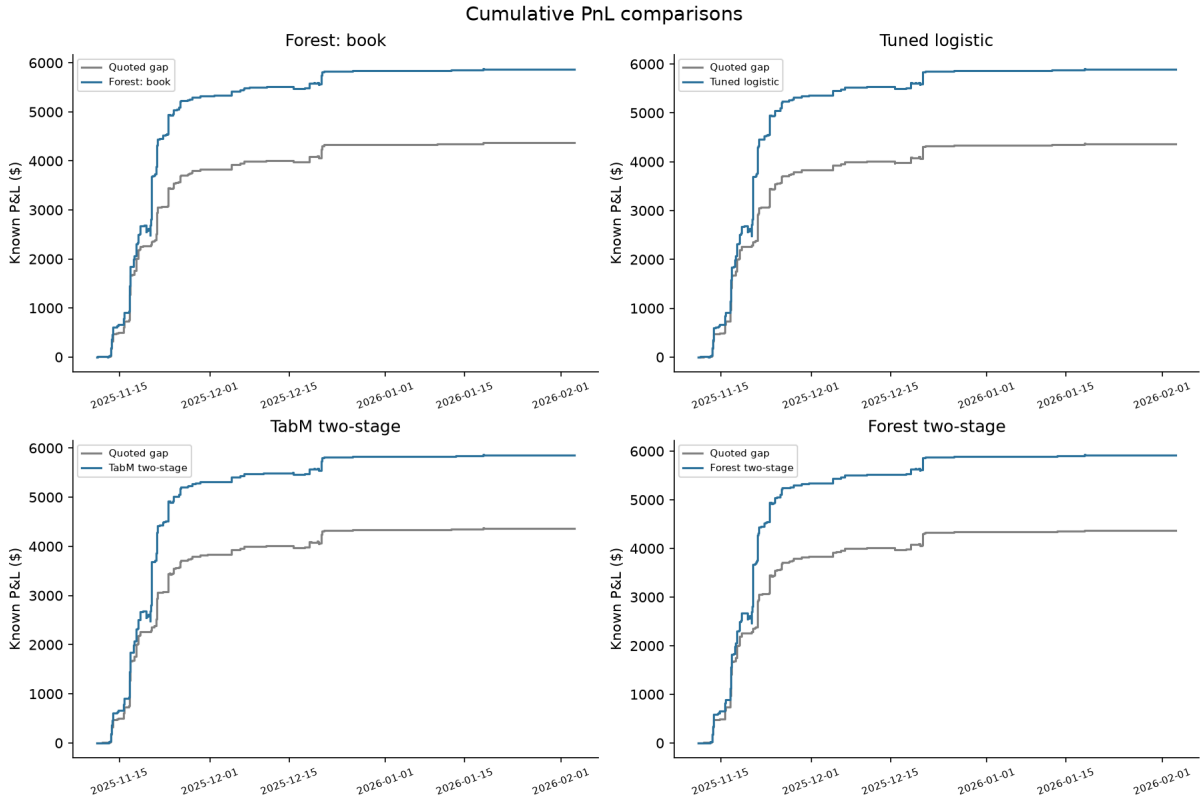


Figure 12. Cumulative PnL comparisons. N=3, 100 ms; cumulative known simulated P&L, with quoted gap repeated as a reference in each panel.

Table 28. December Fed exclusion: reranked N=3, 100 ms.

Policy after event exclusion	**Known P&L ($)**
Forest: book	730.52
Quoted gap	736.36

Without December Fed, reranked forest P&L is $730.52 against quoted gap's $736.36. The economic advantage disappears even though capture remains stronger. The full-cohort gain of $1,503.27 has a weekly interval of $7.93--$3,866.59, but this conditional positive interval does not remove dependence on one event, policy selection or uncertain fills.

7.15 Effect of execution delay

Longer assumed delays reduce what the same monitored events earn. Forest P&L falls from $6,471.72 at zero delay to $5,873.32 at 100 ms and $4,855.08 at 500 ms. Partial-basket results are included in these totals.

Table 29. Selected forest: fixed selections across latency scenarios.

Delay (ms)	Known P&L ($)	Partial-basket P&L ($)
0	6,471.72	0.00
50	6,345.97	-105.11
100	5,873.32	-416.76
250	5,132.97	-1,014.67
500	4,855.08	-1,190.38

Zero delay has no partial loss. At 500 ms, partial outcomes contribute -$1,190.38. A later arrival can lose enough depth to leave an incomplete basket that must be sold at a loss. Keeping the predictor fixed exposes this separate execution dependency. Source-time resolution, competing traders and exchange queues remain outside the model, so these are scenario results rather than evidence of obtainable historical trading profits.

8 Discussion and conclusion

8.1 Answers to the research questions

For RQ1, learning helps allocate monitoring in this cohort. Forest improves substantially over current quoted gap, and current margin alone does not explain the full result. Against historical mean, the pooled value improvement is only 1.12 percentage points. The stronger benefit is retaining more positive hours and serving more events well under equal weighting. Metadata, tuning, ensembles and TabM give mixed improvements. Added complexity therefore needs evidence against strong simple controls.

For RQ2, the preferred policy depends on the question and constraint. More slots reduce the importance of ranking; common token caps can favour logistic or a historical control. Equal-event intervals give a less decisive forest advantage than pooled dollars suggest. Central-bank transfer remains strong, while election transfer weakens. Historical smoothing choices change little; shortening forest training to two weeks matters more. The earlier-only chooser selects forest, but its later evaluation has little remaining dollar value.

For RQ3, retaining opportunities is insufficient to establish economic advantage. Delays create partial losses, and the replay's large positions require substantial funding under its assumptions. Excluding December Fed preserves forest's capture gain over quoted gap but removes its simulated P&L gain. Scoring time quantifies one system cost; the operational saving still depends on unmeasured scanning and monitoring work.

Table 30. Research questions, supporting evidence and the scope of the conclusion.

Question	Evidence	Supported scope
Does learned screening improve capture?	N=3 pooled and event-balanced comparisons, historical controls and paired intervals.	Strong cohort-specific capture; advantage depends on the comparator and weighting.
Which information supports the result?	Controlled feature ablations, family exclusions and paired decision cases.	Predictive dependence within the recorded cohort; no causal attribution.
Does capture survive a common resource cap?	At 50 tokens, forest recall 98.61% and historical-mean recall 96.66%; logistic leads this cap.	Common cap on a proxy; no measured end-to-end efficiency gain.
Can the policy be chosen before its evaluation week?	The earlier-only chooser selects the forest at all nine origins and captures 75/77 positives.	Post-warm-up inspected-data test with full feedback; no untouched final test.
Does opportunity capture establish economic advantage?	Separate simulated execution, latency sensitivity and December Fed exclusion.	Economic advantage over quoted gap disappears on exclusion; no historical-control replay.

8.2 Limits of the evidence

The first limitation is the amount of independent evidence. Only 12 events in eight families have known positive main-evaluation hours, five supported events are ladders, and one event dominates value. Repeated hourly rows do not create independent markets. Reweighting, exclusions and resampling expose this concentration but cannot replace new observations. Policy recommendation and the additional analyses were designed using an already inspected dataset.

Data acquisition materially constrained the study. After Dome's API retired, the original source could no longer expand the frozen universe, improve ladder coverage or supply missing periods. A few recorded events therefore carry much of the conclusion. This does not establish that comparable opportunities are absent elsewhere; broader validation requires a replacement depth archive or prospective collection.

Coverage is a second limitation. Known-only metrics cannot reveal missed opportunities in unknown windows. Frozen membership, metadata and offline recording extent differ from a live catalogue. The processed-data checks reproduce inherited labels without independently auditing raw reconstruction. Training and historical comparisons use recorded earlier

outcomes; acquiring feedback for unselected events in deployment remains unresolved.

The cost and execution models are also incomplete. Token observations ignore different update rates and can count shared tokens twice. Greedy allocation is not optimal, and cached-feature timing covers only four to five candidates on one machine. Replay assumes source-time observation, separate per-leg FOK support, zero fees and extra charges, unlimited funding and a one-second unwind. Queues, competing consumption, network congestion, exchange acknowledgements and settlement cash release are unmodelled. Conditional intervals exclude these uncertainties and are unadjusted for all model comparisons. Selected decision cases illustrate behaviour without identifying causes.

8.3 Next steps

The first priority is a validated replacement for Dome: full historical depth updates or prospective collection that distinguishes missing data from unchanged books. Prices alone cannot establish supported quantities. A broader universe, especially ladders, should be audited and fixed before evaluation, with membership and source/receive timestamps recorded as known at the time. The selected policy then needs independent future events and periods.

Operational work should measure scan construction, update rates, order latency, exchange responses and fills. Learning from selected events should use the labels those selections can actually supply. Fees, bounded capital and settlement release need declared scenarios. New horizons or baskets require rebuilt targets and explicit payoff and order rules. Further tuning should answer a stated uncertainty rather than substitute for better-observed data.

8.4 Conclusion

This dissertation shows how to evaluate limited monitoring separately from execution. Its contribution is a reproducible formulation and implementation with comparisons against current-book and historical priorities. Book-based learning retains strong opportunity capture in the frozen Dome cohort. Its incremental benefit depends on the objective, available history and monitoring budget, while concentration and execution assumptions qualify the economic result. Broader deployment claims require independent data and operational measurements.

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Appendix A Reproduction and artifact guide

The standalone FYP package separates data preparation, structural rules, models, experiments, economic replay and reporting. The notebook research.ipynb orchestrates these stages. Its cached mode verifies saved data and fitted scores and regenerates analyses; its full mode reconstructs examples from raw books, retunes and refits models, and reruns persistent execution into a new run directory. Both modes require the relevant saved local files and operate without a parent repository or provider token. With the Dome API retired, preserving and sharing these files is essential for reproduction.

The data and simulator guides document schemas and assumptions. The artifact table and figure indexes map every catalogued output to a CSV source or supporting dataset. The main report presents selected comparisons; the full 50-figure catalog supports additional inspection without filling the volume with redundant plots.

The processed-data extension is stored separately under artifacts/extensions/processed_review_v1. Its figures, tables, ablation predictions, per-fit audits, input digests and source-backed design record supplement the original 50-figure catalogue. It uses the local earlier opportunity-value examples and saved main-study predictions. The experiment can be rerun with: python FYP/experiments/processed_extension.py --examples PATH_TO_EARLIER_EXAMPLES. The extension neither accesses Dome nor reconstructs raw books. Its original-study results are checked against the saved N=3 and N=5 metrics before new analyses proceed.

The additional robustness extension is stored under artifacts/extensions/robustness_review_v1. Running python FYP/experiments/robustness_extension.py reproduces its four historical controls, 252 family-withheld fits and balanced metrics from the packaged processed examples and saved predictions. A source snapshot, frozen design, input fingerprints, per-scan history cutoffs, per-fit family exclusions, predictions and paired interval outputs accompany the results. Every family scenario scores the whole eligible pool before its target-family outcomes are summarised. These outputs add two figures to the separate extension catalogues.

The extension includes a compact copy of its earlier processed examples under its inputs directory. The full raw-book archive and complete main-study feature table are required for the original end-to-end reconstruction but are not used by this extension. Running python rebuild_report.py regenerates the presentation tables and dissertation from saved evidence without fitting models or accessing raw books. All new ablation labels inherit the original preprocessing assumptions and have not been independently re-established from raw snapshots in this revision.

The allocation extension under artifacts/extensions/allocation_review_v1 uses the same saved predictions, earlier examples and historical-control scores. Running python FYP/experiments/allocation_extension.py reproduces token-cap allocations, the four-week-warm-up chooser and paired cases without model refitting. Its design and executed source accompany per-scan budget audits, per-origin selection histories, complete case features and competing candidate tables. Its resource curve adds one figure to the extension catalogues.

The 28 tests exercise payoff constraints, backward as-of reads, duplicate timestamps, recording endpoints, empty versus unknown books, integer depth sizing, uncapped quantities, tick/minimum constraints, short opportunities, incomplete labels, half-open windows, peak-value aggregation, positive-only amount fitting, unknown slot consumption, chronological purging, selection invariance, rank ensembles, primary execution settings, persistent reservations and partial/unknown execution outcomes. They test implemented safeguards, not the completeness of the original collector or the truth of real fills.

Three additional tests under extension_tests check strict historical-label availability, invariance to changes in later labels, and exclusion of unknown windows from historical counts. They also check that a known zero replaces an earlier positive as the last observed value. These checks protect the new historical controls without changing the original 28-test suite.

Five allocation tests additionally check that expensive events are skipped without exceeding the cap, unknown outcomes retain their resource allocation, labels ending exactly at a chooser origin remain unavailable, changes to later targets cannot alter policy choice, and illustrative cases use disjoint groups of known positives. Two further rolling-window tests verify the included lower scan boundary, the excluded label-end boundary, unknown-label exclusion and nested available histories. The extension suites contain ten tests in addition to the original 28.

The focused sensitivity study under artifacts/extensions/sensitivity_review_v1 is reproduced with python FYP/experiments/sensitivity_extension.py. It records the three smoothing strengths, two rolling windows, 54 new fits and the reused expanded controls in a frozen design and source snapshot. Saved predictions, availability and fit audits, matched-period metrics and original-score reconciliations support each comparison. Timing samples include every measured call, scan size, avoided token count, timer unit, environment and declared exclusions. Forest timing configurations preserve the saved scores. Warm timings are retained on resume; rerunning on another machine requires a new output directory. Presentation tables are regenerated from these saved measurements, not retimed during a report build.

Appendix B Configuration and provenance

inputs/cohort.json records complete basket membership and validity intervals. inputs/raw_inventory.csv records every required raw history and its digest, a file fingerprint used to detect changes. inputs/study.json records the feature views and bounded configuration grids. Weekly bundle files record selected parameters and feature order; fit audits record the actual training populations and cutoffs. The search contains 168 inner candidate evaluations, and the weekly audit records 199 distinct fitted artifacts or stages. Identical fixed and tuned classifier configurations can share an artifact, so this count does not represent independent supporting experiments. Tuning search and selected hyperparameter CSVs provide the detailed configurations supporting Chapter 5.

Source, input and output digests detect changes during reproduction. Verification recomputes weekly scores from fitted artifacts, checks temporal separation, reconstructs capture metrics and reconciles cumulative known P&L with economic summaries. These controls establish reproducibility of the supplied study, not point-in-time availability of frozen metadata or completeness of the feed.

Appendix C Exact search configurations

The following tables enumerate the declared candidate lists used by the primary study. IDs are zero-based and resolve otherwise exact tuning ties by declaration order. Fixed classical controls use candidate zero; the two-stage forest uses the separate settings in Table 7. TabM lists the four configurations actually searched, rather than the larger upstream architecture catalogue. The selected settings and refit durations are recorded below so a reader need not reconstruct them from serialised model files.

Table 31. Logistic regression candidate configurations; zero-based IDs.

ID	Configuration
0	C=1.0; penalty=l2; class_weight=balanced; solver=lbfgs
1	C=0.01; penalty=l2; class_weight=balanced; solver=lbfgs
2	C=0.1; penalty=l2; class_weight=balanced; solver=lbfgs
3	C=10.0; penalty=l2; class_weight=balanced; solver=lbfgs
4	C=0.1; penalty=l1; class_weight=balanced; solver=liblinear
5	C=1.0; penalty=l1; class_weight=balanced; solver=liblinear
6	C=0.1; penalty=l2; class_weight=None; solver=lbfgs
7	C=1.0; penalty=l2; class_weight=None; solver=lbfgs

Table 32. Random forest candidate configurations; zero-based IDs.

ID	Configuration
0	n_estimators=200; max_depth=10; min_samples_leaf=20; max_features=sqrt
1	n_estimators=200; max_depth=4; min_samples_leaf=20; max_features=0.5
2	n_estimators=200; max_depth=8; min_samples_leaf=5; max_features=0.5
3	n_estimators=200; max_depth=8; min_samples_leaf=50; max_features=1.0
4	n_estimators=200; max_depth=12; min_samples_leaf=20; max_features=0.5
5	n_estimators=200; max_depth=None; min_samples_leaf=20; max_features=0.5
6	n_estimators=200; max_depth=6; min_samples_leaf=50; max_features=0.5
7	n_estimators=200; max_depth=12; min_samples_leaf=5; max_features=1.0

Table 33. XGBoost candidate configurations; zero-based IDs.

ID	Configuration
0	n_estimators=200; learning_rate=0.05; max_depth=3; min_child_weight=10; subsample=1.0; colsample_bytree=1.0; reg_lambda=10.0; reg_alpha=0.0
1	n_estimators=200; learning_rate=0.05; max_depth=2; min_child_weight=10; subsample=0.8; colsample_bytree=0.8; reg_lambda=10.0; reg_alpha=0.0
2	n_estimators=400; learning_rate=0.03; max_depth=3; min_child_weight=20; subsample=0.8; colsample_bytree=0.8; reg_lambda=10.0; reg_alpha=0.0
3	n_estimators=200; learning_rate=0.05; max_depth=4; min_child_weight=20; subsample=0.8; colsample_bytree=0.8; reg_lambda=10.0; reg_alpha=1.0
4	n_estimators=200; learning_rate=0.05; max_depth=6; min_child_weight=20; subsample=0.8; colsample_bytree=0.8; reg_lambda=10.0; reg_alpha=1.0
5	n_estimators=400; learning_rate=0.03; max_depth=2; min_child_weight=10; subsample=1.0; colsample_bytree=1.0; reg_lambda=1.0; reg_alpha=0.0
6	n_estimators=200; learning_rate=0.1; max_depth=3; min_child_weight=20; subsample=0.8; colsample_bytree=0.8; reg_lambda=100.0; reg_alpha=1.0
7	n_estimators=200; learning_rate=0.05; max_depth=3; min_child_weight=50; subsample=1.0; colsample_bytree=0.8; reg_lambda=100.0; reg_alpha=10.0

Table 34. Two-stage TabM candidate configurations; zero-based IDs.

ID	Configuration
0	n_blocks=2; d_block=128; dropout=0.1; lr=0.002; weight_decay=0.0003
1	n_blocks=1; d_block=64; dropout=0.0; lr=0.002; weight_decay=0.0003
2	n_blocks=2; d_block=64; dropout=0.1; lr=0.001; weight_decay=0.001
3	n_blocks=3; d_block=128; dropout=0.1; lr=0.001; weight_decay=0.0003

Table 35. Selected configurations at the three tuning origins.

Origin	Family	ID	Selected refit epochs
week_00	logistic	6	Not applicable
week_00	random forest	2	Not applicable
week_00	xgboost	1	Not applicable
week_00	tabm two stage	0	{"classifier": 38, "amount": 21}
week_04	logistic	7	Not applicable
week_04	random forest	0	Not applicable
week_04	xgboost	5	Not applicable
week_04	tabm two stage	2	{"classifier": 57, "amount": 31}
week_08	logistic	7	Not applicable
week_08	random forest	0	Not applicable
week_08	xgboost	4	Not applicable
week_08	tabm two stage	1	{"classifier": 58, "amount": 8}

Appendix D Implementation details

This appendix retains the numerical preprocessing, optimisation and validation details supporting Chapter 5. They are part of the implemented protocol; moving them here leaves the model fits and reported results unchanged.

D.1 Classical preprocessing and training

The classifiers optimise their standard classification losses rather than a differentiable top-N ranking loss. Logistic regression uses penalised binary log loss; the forest grows Gini-criterion trees; XGBoost uses the binary logistic objective with histogram tree construction. The fixed classifiers use class balancing, including a training-count negative/positive ratio for XGBoost. The two-stage gates deliberately omit this balancing because their outputs multiply a conditional value estimate. All classical models use the declared seed 20261001 and four worker threads where applicable.

For the classical models, missing numeric values are replaced with training-set medians, with additional indicators identifying missing values where applicable. A signed logarithmic transform compresses large magnitudes while retaining their signs, followed by standardisation. Categorical metadata is represented by binary indicators for categories observed in training; unseen categories introduce no new indicators at prediction time. All preprocessing is fitted on the relevant training rows and saved with the model, so validation and evaluation data do not determine medians, scales or category mappings.

The signed transform is $\operatorname{sgn}(x)\log(1+|x|)$. Standardisation uses training-fitted means and variances after imputation and transformation. An entirely missing numeric feature is retained by the imputer, preserving the declared input schema. The metadata comparison adds event type, broad topic and creation age; it does not introduce event identifiers as category features. The numeric age follows the same classical pipeline as book measurements.

Each tuned classical family evaluates eight hyperparameter configurations. Logistic settings vary regularisation strength, L1/L2 penalty and class weighting. Forest settings vary depth, leaf size and feature subsampling with 200 trees. XGBoost settings vary tree count, depth, learning rate, child weight, subsampling and regularisation. These bounded searches compare declared alternatives rather than exhaustively optimise each family; the exact grids and selected settings are exported.

D.2 TabM architecture, losses and refits

TabM uses 32 ensemble members within each stage and tests four configurations from the declared architecture grid. Continuous variables use a training-fitted quantile transformation to a normal distribution, and categories use one-hot encoding. Training computes losses separately for the members; prediction averages their probabilities or amount estimates. Batch size is 1,024, and training runs for at most 60 epochs, where an epoch is one pass through the training examples. Early stopping allows ten epochs without an improvement in validation loss.

TabM uses the official tabm architecture with K=32 members and one output per member. The four candidates have block/width/dropout settings (2,128,0.1), (1,64,0), (2,64,0.1) and (3,128,0.1); their optimiser settings are given in Appendix C. AdamW updates the model, with gradient norm clipped to one. Continuous features receive median imputation and a normal-output quantile transform fitted with at most 1,000 quantiles and no subsampling. Binary numeric columns use most-frequent imputation; categorical columns use one-hot encoding.

For the probability stage, each member receives the binary label and contributes a binary cross-entropy loss computed from its logit. For the amount stage, positive target values are standardised by their training mean and standard deviation, with scale bounded below by one millionth. Each member contributes squared error on this standardised target. The batch losses average over observations and members. Inference averages member probabilities for the gate and member values for the amount model, reverses the amount scaling, and clips negative amount predictions to zero.

Early stopping is assessed on the ensemble prediction: binary cross-entropy of mean gate probabilities, or squared error of mean standardised amount predictions. The validation examples fit neither feature preprocessing nor target scaling. For a fixed-epoch weekly refit, no validation outcomes are supplied. The three refit seeds are 20261001, 20261002 and 20261003; averaging occurs after multiplying each seed-specific gate and amount prediction.

Inner tuning uses one fixed random seed. Weekly refits train three seed-specific pairs of probability and amount models, multiply the two predictions within each pair, then average the three products. For each stage, the refit duration is the median best epoch count from the selected configuration's inner validation fits, rounded up. This duration is retained until the next scheduled tuning date, so subsequent evaluation outcomes do not determine when training stops.

D.3 Exact validation boundaries

The inner boundaries are defined from distinct scheduled scan times in the earlier history. The first block begins at the halfway indexed scan, the second at the three-quarter indexed scan, and the second ends one hour before the outer fitting origin. Because the first recorded scan is not midnight, these inner boundaries need not align with midnight or whole calendar weeks. Table 8 records their actual timestamps. Every candidate for a family sees the same validation candidate sets, including unknown rows that consume slots. Known labels alone supply fitting targets and capture denominators.

D.4 Historical estimates and policy-choice ties

The controls rank by past event opportunity frequency, past event mean quoted value, the event's last known quoted value, or past family mean quoted value. For frequency and mean value, a family estimate is its observed sum plus 24 times the global observed mean, divided by its known count plus 24. An event estimate uses the same calculation with the corresponding family estimate as its prior. The fixed 24-observation smoothing strength represents one day's hourly observations and is not tuned. An event without history falls back to its family estimate; a globally empty history gives zero. Last-known value includes observed zeros and uses family mean value only when that event has no known history. Ties retain the main study's event-identifier ordering.

Earlier-only policy choice treats differences within $0.000000001 as ties and uses the policy order in Section 5.7. This is a numerical tolerance for the choice procedure, not a profit threshold. Individual scores, selected rows and cutoff audits are retained in the allocation artifacts described in Appendix A.

B.Comp. Dissertation

AI in financial data analytics and trading

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